Data Handling: Group Project

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Deadline: Please submit your homework by November 6th, 2025, 23:59 (part 1) and by December 10th, 2025, 23:59 (part 2).

Format: A rendered Quarto HTML document using the official course template available on Canvas.

Main text: Use the main text to write your answers to the questions. Be concise in your answers.

Code: Use code chunks to write and run your code. All code chunks must be visible in the document. The code must be spaced and readable (for inspiration, see https://adv-r.had.co.nz/Style.html)

Ensure that:

- All variables are defined in the environment,
- Any required code (e.g., functions) is included in the document.

Important: We must be able to compile and render your Quarto document without errors.

For any questions, please use the discussion platform on Canvas. Questions by email will not be answered.

0.1 R Setup

```r
# Load necessary libraries
library(httr)
library(httr2)
library(jsonlite)
library(dplyr)
library(tidyr)
```

Homework Part 1 (due November 6th, 2025)

The relationship between health expenditures and health outcomes has long been a central topic in health economics. Empirical studies show a positive but non-linear relationship between health spending and outcomes such as life expectancy, particularly in low- and middle-income countries.

In high-income settings, however, diminishing returns are evident. For example, the United States spends more per capita on health than any other nation, yet its health outcomes often lag behind those of countries with lower spending (Anderson et al. (2003)).

In this group project, we investigate the relationship between health expenditures and health outcomes.

1.1 Importing data on healthcare expenditures per capita

Retrieve and import data on healthcare expenditures per capita from the World Bank. The indicators to be imported are[1]:

- *Current health expenditure per capita (in current US$)* (Indicator WB_WDI_SH_XPD_CHEX_PC_CD), for the years 2000 to 2022.[2]
- *Current health expenditure (% of GDP)* (Indicator SH_XPD_CHEX_GD_ZS) for years 2000 to 2023.

1.1.1 Current health expenditure per capita and current health expenditure (% of GDP)

Use your favorite method and present **one** solution. We explored different approaches to using APIs in Minna Heim's guest lecture. Explain the method of your choice in your own words and justify why you chose it.[5pts]

Hint: there are different methods to import data from the World Bank website. For example:

- *data360.worldbank.org API*.
- the older World Bank API

[1] More information on these two indicators can be accessed here and here.
[2] Note that healthcare spending varies widely between countries, and that the indicator in Purchasing Power Parity (PPP) "Current health expenditure per capita, PPP" would also be a good indicator to compare countries over time. We use the indicator in current US dollars to simplify our analysis.

- R-friendly API wrappers (packages) that simplify access to World Bank data.

1.1.1.1 Solution 1: Use WDI package

To provide a correct solution, many different methods were available. The easiest solution was to use the API wrapper. There are (at least) two wrappers for World Bank data:

- [WDI](https://github.com/vincentarelbundock/WDI)
- [wbstats](https://github.com/gshs-ornl/wbstats)

Other solutions are also correct. Both tables can be imported right away, and in one single table. Or we can import both of them separately.

```
#------------------------------------------------------------------------------
# Solution 1: WDI package all indicators at once
#------------------------------------------------------------------------------
library(WDI) #source: https://github.com/vincentarelbundock/WDI

# Search for "health expenditure" indicators
indicators <- WDIsearch("health expenditure")
head(indicators, 10)  # View first 10 results
```

```
                   indicator
10155       HOU.XPD.HE.PC.CR
10209 hxpd_al_alot_dfcl_1529
10210 hxpd_al_alot_dfcl_3044
10211 hxpd_al_alot_dfcl_4564
10212 hxpd_al_alot_dfcl_65up
10213  hxpd_al_alot_dfcl_all
10214  hxpd_al_alot_dfcl_fem
10215 hxpd_al_alot_dfcl_male
10216  hxpd_al_alot_dfcl_rur
10217  hxpd_al_alot_dfcl_urb
                                                                            name
10155                                        Monthly Per Capita Household Health
Expenditure (in IDR)
10209   Household health expenditures (% of adults aged 15 to 29 years with
severe disability)
10210   Household health expenditures (% of adults aged 30 to 44 years with
severe disability)
10211   Household health expenditures (% of adults aged 45 to 64 years with
severe disability)
10212      Household health expenditures (% of adults aged 65 or older with
severe disability)
10213                            Household health expenditures (% of adults with
severe disability)
10214                     Household health expenditures (% of female adults with
severe disability)
10215                       Household health expenditures (% of male adults with
```

```
severe disability)
10216 Household health expenditures (% of adults living in rural areas with
severe disability)
10217 Household health expenditures (% of adults living in urban areas with
severe disability)
```

```
df_WorldBank_health <- WDI(
  country = "all",
  indicator = c(
    "ind_WorldBank_health_gdp" = "SH.XPD.CHEX.GD.ZS", # How important health
spending is in the overall economy
     "ind_WorldBank_health_usd" = "SH.XPD.CHEX.PC.CD" # How much is spent on
health per person

  ),
  start = 2000,
  end = 2022,
  extra = TRUE  # Adds region, income, etc.
)


#-------------------------------------------------------------------------------
# Solution 1: WDI package each indicator separately
#-------------------------------------------------------------------------------
library(WDI)

# Search for "health expenditure" indicators
indicators <- WDIsearch("health expenditure")
head(indicators, 10)  # View first 10 results
```

```
                   indicator
10155       HOU.XPD.HE.PC.CR
10209 hxpd_al_alot_dfcl_1529
10210 hxpd_al_alot_dfcl_3044
10211 hxpd_al_alot_dfcl_4564
10212 hxpd_al_alot_dfcl_65up
10213  hxpd_al_alot_dfcl_all
10214  hxpd_al_alot_dfcl_fem
10215 hxpd_al_alot_dfcl_male
10216  hxpd_al_alot_dfcl_rur
10217  hxpd_al_alot_dfcl_urb
                                                                          name
10155                                  Monthly Per Capita Household Health
Expenditure (in IDR)
10209   Household health expenditures (% of adults aged 15 to 29 years with
severe disability)
10210   Household health expenditures (% of adults aged 30 to 44 years with
severe disability)
10211   Household health expenditures (% of adults aged 45 to 64 years with
```

```
severe disability)
10212      Household health expenditures (% of adults aged 65 or older with
severe disability)
10213                       Household health expenditures (% of adults with
severe disability)
10214                Household health expenditures (% of female adults with
severe disability)
10215                  Household health expenditures (% of male adults with
severe disability)
10216 Household health expenditures (% of adults living in rural areas with
severe disability)
10217 Household health expenditures (% of adults living in urban areas with
severe disability)
```

```
df_WorldBank_health_gdp <- WDI(
  country = "all",
  indicator = "SH.XPD.CHEX.GD.ZS",
  start = 2000,
  end = 2022,
  extra = TRUE
)


df_WorldBank_health_usd <- WDI(
  country = "all",
  indicator = "SH.XPD.CHEX.PC.CD",
  start = 2000,
  end = 2022,
  extra = TRUE
)
```

1.1.1.2 Solution 2: Use the traditional API of the World Bank

For this particular API, I used Minna Heim's function we saw during the course. From the API documentation, the url needs an indicator, a start and end date, and the number of rows we want to retrieve. It renders a json, which we parsed into a list using `resp_body_json()`.

```r
#----------------------------------------------------------------------------------
# Solution 2: traditional API
#----------------------------------------------------------------------------------

# Retrieve Minna Heim's function
get_API_wb <- function(indicator, start_year, end_year, per_page = 2000){
  base_url <- "http://api.worldbank.org/v2/country/all/indicator/"

  final_url <- paste0(
    base_url,
    indicator,
    "?format=json&date=",
    start_year,
    ":",
    end_year,
    "&per_page=",
    per_page
  )

  # perform API call with httr2
  req <- request(final_url)
  resp <- req_perform(req)

  # if request is not successful
  if (resp_status(resp) != 200){
    message("The request was not successful")
  }
  else{
    return(resp_body_json(resp, simplifyVector = T))
    # I was not happy with the way it parsed the json, so I checked the
    # documentation for resp_body_json. I want all arrays in the json to
    # be caused to atomic vectors, so simplifyVector = T
  }
}

# Initiate an empty list to store results from the loop.
list_container <- list()

# List of indicators
indicators <- c("SH.XPD.CHEX.GD.ZS", "SH.XPD.CHEX.PC.CD")

# Start loop over the different indicators
for (ii in indicators){
  print(ii)
```

```r
  json_data <- get_API_wb(indicator = ii, start_year = 2000, end_year = 2022,
                          per_page = 10000)


  # Extract and convert to data frame
  table <- json_data[[2]]
  str(table)

  # Weird: indicator and country are dataframes within dataframes
  table$indicator

  # Extract and clean data
  # Step 1: Unnest 'indicator'
  table$indicator_id <- table$indicator$id
  table$indicator_value <- table$indicator$value

  # Step 2: Unnest 'country'
  table$country_id <- table$country$id
  table$country_name <- table$country$name

  # Step 3: Drop the original nested columns
  table$indicator <- NULL
  table$country <- NULL

  list_container[[ii]] <- table
}
```

```
[1] "SH.XPD.CHEX.GD.ZS"
'data.frame':   6118 obs. of  8 variables:
 $ indicator      :'data.frame':    6118 obs. of  2 variables:
  ..$ id   : chr  "SH.XPD.CHEX.GD.ZS" "SH.XPD.CHEX.GD.ZS" "SH.XPD.CHEX.GD.ZS"
"SH.XPD.CHEX.GD.ZS" ...
  ..$ value: chr  "Current health expenditure (% of GDP)" "Current health
expenditure (% of GDP)" "Current health expenditure (% of GDP)" "Current
health expenditure (% of GDP)" ...
 $ country        :'data.frame':    6118 obs. of  2 variables:
  ..$ id   : chr  "ZH" "ZH" "ZH" "ZH" ...
  ..$ value: chr  "Africa Eastern and Southern" "Africa Eastern and Southern"
"Africa Eastern and Southern" "Africa Eastern and Southern" ...
 $ countryiso3code: chr  "AFE" "AFE" "AFE" "AFE" ...
 $ date           : chr  "2022" "2021" "2020" "2019" ...
 $ value          : num  5.66 5.97 6.02 5.78 5.76 ...
 $ unit           : chr  "" "" "" "" ...
 $ obs_status     : chr  "" "" "" "" ...
 $ decimal        : int  2 2 2 2 2 2 2 2 2 2 ...
[1] "SH.XPD.CHEX.PC.CD"
'data.frame':   6118 obs. of  8 variables:
 $ indicator      :'data.frame':    6118 obs. of  2 variables:
  ..$ id   : chr  "SH.XPD.CHEX.PC.CD" "SH.XPD.CHEX.PC.CD" "SH.XPD.CHEX.PC.CD"
```

```
"SH.XPD.CHEX.PC.CD" ...
  ..$ value: chr  "Current health expenditure per capita (current US$)" "Current
health expenditure per capita (current US$)" "Current health expenditure
per capita (current US$)" "Current health expenditure per capita (current
US$)" ...
 $ country        :'data.frame':    6118 obs. of  2 variables:
  ..$ id   : chr  "ZH" "ZH" "ZH" "ZH" ...
  ..$ value: chr  "Africa Eastern and Southern" "Africa Eastern and Southern"
"Africa Eastern and Southern" "Africa Eastern and Southern" ...
 $ countryiso3code: chr  "AFE" "AFE" "AFE" "AFE" ...
 $ date           : chr  "2022" "2021" "2020" "2019" ...
 $ value          : num  92.6 92.4 81.6 87.3 90.1 ...
 $ unit           : chr  "" "" "" "" ...
 $ obs_status     : chr  "" "" "" "" ...
 $ decimal        : int  2 2 2 2 2 2 2 2 2 2 ...
```

```
# Preview
df_WorldBank_health_gdp <- list_container$SH.XPD.CHEX.GD.ZS
df_WorldBank_health_usd <- list_container$SH.XPD.CHEX.PC.CD
```

1.1.1.3 Solution 3: Use API360

The API 360 is generally more difficult to use, poorly documented, and delivers less observations per request than the other two solutions. It is only possible to retrieve 1000 rows at a time. Therefore, a loop must be programmed.

```
#-------------------------------------------------------------------------------
# Solution 3:  API old version discontinued
# https://data360.worldbank.org/en/api?indicatorid=WB_WDI_SH_XPD_CHEX_PP_CD&
datasetid=WB_WDI
#-------------------------------------------------------------------------------

# Define API URL: check "execute" on the data360 API webpage
# Number of records to skip. Each API call returns a maximum of 1000 records.
# Use the skip parameter to paginate through the results.

list_df <- list()

for (jj in seq(0, 5000, by = 1000)) {

  url <- paste0(
          "https://data360api.worldbank.org/data360/data?DATABASE_ID=WB_WDI&
INDICATOR=WB_WDI_SH_XPD_CHEX_PC_CD",
    "&timePeriodFrom=2000&timePeriodTo=2022&skip=", jj
  )

  # Send GET request
  response <- GET(url)

  # Parse JSON content
  dat <- content(response, as = "parsed", simplifyVector = TRUE)

  # Extract relevant data block
  list_df[[jj/1000+1]] <- dat$value

}

data_api360 <- bind_rows(list_df)

data_api360 <- data_api360 |>
  select(
    countryiso3code = REF_AREA,
    year =  TIME_PERIOD,
    value = OBS_VALUE
  )
```

1.1.2 What are the healthcare expenditures for Albania in 2005 in % of GDP?

Explore your dataset and report the healthcare expenditures for Albania in 2005, expressed as a percentage of GDP. Round the value to two decimal places. [1 pt]

1.1.2.1 Solution

Any solution is fine, as long as it retrieves the value 5.73 for Albania in 2005.

One possible answer using the dplyr syntax is given below. Note that we have not seen dplyr in the lecture yet.

```
df_WorldBank_health  |>
  filter(country == "Albania", year == 2005)  |>
  select(ind_WorldBank_health_gdp) |>
  round(2) # 5.73
```

```
  ind_WorldBank_health_gdp
1                     5.73
```

```
df_WorldBank_health_gdp |>
  filter(countryiso3code == "ALB" & date == 2005) |>
  select(value) |>
  round(2) # 5.73
```

```
  value
1  5.73
```

1.1.3 Plausibility checks

Are these data similar to the WHO data on health expenditures? Have a look at the WHO website.

- Why is it important to compare the same indicators from different sources? [1 pt]
- Use the WHO Data Explorer to find health expenditures in Albania in 2005. Are the results similar to your findings? [1 pt]

1.1.3.1 Solution

I simply went on the visualization page for Albania (here: https://apps.who.int/nha/database/country_profile/Index/en). I could confirm that both the ind_WorldBank_health_usd and the ind_WorldBank_health_gdp are similar.

I also checked with their Data explorer:

- Indicators and data: INDICATORS/AGGREGATES/CURRENT HEALTH EXPENDITURE as % GDP
- Country: Albania
- Year: 2005
- Units of Expenditure: %GDP

```
df_WorldBank_health  |>
  filter(country == "Albania", year == 2005)  |>
  select(ind_WorldBank_health_usd) |>
  round(2) # 149.98
```

```
  ind_WorldBank_health_usd
1                   149.98
```

1.2 Importing data on life expectancy and perceived health status for OECD countries

Our research question examines the relationship between health outcomes and investment in health. In the previous tasks, we retrieved data on investment in health. We now focus on retrieving data on life expectancy and perceived health status.

1.2.1 Life expectancy at birth indicator

Import the life expectancy at birth indicator from the World Bank. Use your preferred method and present **one** solution. Explain the method in your own words and justify why you chose it. [3 pts]

1.2.1.1 Solution

There are different possible solutions: use the API like in the first exercise, use the WDI package, or simply download and load a csv file.

```
#-----------------------------------------------------------------------------
# Importing data on life expectancy for OECD countries
#-----------------------------------------------------------------------------

# Check using the WDI package
WDIsearch('expecta.')
```

```
                   indicator
18386      SABER.TECH.GOAL1
18387 SABER.TECH.GOAL1.LVL1
22084      SP.DYN.LE00.FE.IN
22085         SP.DYN.LE00.IN
22086      SP.DYN.LE00.MA.IN
22087      SP.DYN.LE60.FE.IN
22088      SP.DYN.LE60.MA.IN
22089         SP.DYN.LIFE.MF
                                                                        name
18386           SABER: (Teachers) Policy Goal 1: Setting clear expectations
for teachers
18387 SABER: (Teachers) Policy Goal 1 Lever 1: Are there clear expectations
for teachers?
22084                                             Life expectancy at birth,
female (years)
22085                                             Life expectancy at birth,
total (years)
22086                                             Life expectancy at birth,
male (years)
22087                                            Life expectancy at age 60,
female (years)
22088                                            Life expectancy at age 60,
```

```
male (years)
22089                                                          Life Expectancy
at Birth(years)
```

```
df_WorldBank_life_expectancy = WDI(indicator='SP.DYN.LE00.IN', start = 2000,
end = 2022)
```

1.2.2 Perceived Health Status

Retrieve the indicators on life expectancy and perceived health status from the OECD Data Explorer. You may need to select specific data before importing, or you can import the full dataset.

Use your preferred method to retrieve the data and, if you made any data selection before importing, explain your reasoning. Present **one** solution. [5 pts]

Hints: See the API explanations to understand how to retrieve data from the OECD Data explorer.

There are different possible solutions to retrieve the data. Of course, using the wrapper is the easiest. There is another solution suggested on the API documentation using `read.csv()`.

```
#------------------------------------------------------------------------------
# Importing data on life expectancy and perceived health status from the
# OECD API.
#------------------------------------------------------------------------------

# Life expectancy
url_le      <-      "https://sdmx.oecd.org/public/rest/data/OECD.ELS.HD,DSD_
HEALTH_STAT@DF_LE,/.A...Y0........?startPeriod=2015&dimensionAtObservation=
AllDimensions&format=csvfilewithlabels&startPeriod=2000"
oecd_le <- read.csv(url_le)
nrow(oecd_le)
```

```
[1] 6120
```

```
# Perceived health status
url_hs      <-      "https://sdmx.oecd.org/public/rest/data/OECD.ELS.HD,DSD_
HEALTH_STAT@DF_HEALTH_STATUS,1.0/.A.PHS..._T.......?dimensionAtObservation=
AllDimensions&format=csvfilewithlabels&startPeriod=2000"
oecd_hs <- read.csv(url_hs)
nrow(oecd_hs)
```

```
[1] 8404
```

```r
# Alternative: Use the wrapper package (easier and more reliable)
# install.packages("OECD")
library(OECD)

dataset_id <- "OECD.ELS.HD,DSD_HEALTH_STAT@DF_HEALTH_STATUS,1.0"

# Fetch the complete health status dataset
df <- get_dataset(dataset_id,
                  start_time = 2000, end_time = 2022)

# Filter for life expectancy at birth
df_oecd_life_expectancy <- df %>%
  filter(MEASURE == "LFEXP", AGE == "Y0")

# Filter for perceived health status
df_oecd_phs <- df %>%
  filter(MEASURE == "PHS")
```

1.3 First data exploration

In his book *Statistics for Public Policy*, Jeremy G. Weber argues that one of the most important steps in data analytics is to know your data. By this, he means understanding key characteristics of the dataset. He highlights three essential aspects:

1. Know the **definitions** (for example, how is perceived health status defined?)
2. Know the **units** (for example, are things measured in euros, swiss francs, or dollars?)
3. Know the **quality** of your data. Conduct simple “smell tests” on key variables.

We can add two additional aspects:

4. Know the **unit of observation** (are the things we are observing countries, groups of countries, continents?)
5. Know the **time structure** (are we observing the same unit repeatedly over time, as in a panel, or different units at different times, as in a cross-section?)

Now explore the data frames you have imported.

1.3.1 Content of your data

Briefly describe the content of each dataset according to the five key aspects described above [8 pts].

1.3.1.1 Solutions

No solution posted for this exercise.

1.3.2 Data dimensions and structure

Report the dimensions of your data frames (number of rows and columns). Comment on their structure and describe the types of variables they contain (for example, character, numeric, factor, date, etc.). [2 pts].

```r
str(oecd_le) # gives you the dimensions, var type and overview
```

```
'data.frame':	6120 obs. of  44 variables:
 $ STRUCTURE                         : chr  "DATAFLOW" "DATAFLOW" "DATAFLOW"
"DATAFLOW" ...
 $ STRUCTURE_ID                 : chr  "OECD.ELS.HD:DSD_HEALTH_STAT@DF_LE(1.1)"
"OECD.ELS.HD:DSD_HEALTH_STAT@DF_LE(1.1)"
"OECD.ELS.HD:DSD_HEALTH_STAT@DF_LE(1.1)"
"OECD.ELS.HD:DSD_HEALTH_STAT@DF_LE(1.1)" ...
 $ STRUCTURE_NAME              : chr  "Life expectancy" "Life expectancy" "Life
```

```
expectancy" "Life expectancy" ...
 $ ACTION                    : chr  "I" "I" "I" "I" ...
 $ REF_AREA                  : chr  "AUT" "BEL" "CHL" "CHL" ...
 $ Reference.area            : chr  "Austria" "Belgium" "Chile" "Chile" ...
 $ FREQ                      : chr  "A" "A" "A" "A" ...
 $ Frequency.of.observation  : chr  "Annual" "Annual" "Annual" "Annual" ...
 $ MEASURE                   : chr  "LFEXPDFM" "LFEXPDFM" "LFEXPDFM"
"LFEXPDFM" ...
 $ Measure                   : chr  "Life expectancy difference (female-
male)" "Life expectancy difference (female-male)" "Life expectancy difference
(female-male)" "Life expectancy difference (female-male)" ...
 $ UNIT_MEASURE              : chr  "Y" "Y" "Y" "Y" ...
 $ Unit.of.measure           : chr  "Years" "Years" "Years" "Years" ...
 $ AGE                       : chr  "Y0" "Y0" "Y0" "Y0" ...
 $ Age                       : chr  "0 years" "0 years" "0 years" "0 years" ...
 $ SEX                       : chr  "_T" "_T" "_T" "_T" ...
 $ Sex                       : chr  "Total" "Total" "Total" "Total" ...
 $ SOCIO_ECON_STATUS         : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Socio.economic.status     : chr  "Not applicable" "Not applicable" "Not
applicable" "Not applicable" ...
 $ DEATH_CAUSE               : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Cause.of.death            : chr  "Not applicable" "Not applicable" "Not
applicable" "Not applicable" ...
 $ CALC_METHODOLOGY          : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Calculation.methodology   : chr  "Not applicable" "Not applicable" "Not
applicable" "Not applicable" ...
 $ GESTATION_THRESHOLD       : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Gestation.period.threshold: chr  "Not applicable" "Not applicable" "Not
applicable" "Not applicable" ...
 $ HEALTH_STATUS             : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Health.status             : chr  "Not applicable" "Not applicable" "Not
applicable" "Not applicable" ...
 $ DISEASE                   : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Disease                   : chr  "Not applicable" "Not applicable" "Not
applicable" "Not applicable" ...
 $ CANCER_SITE               : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Cancer.site               : chr  "Not applicable" "Not applicable" "Not
applicable" "Not applicable" ...
 $ TIME_PERIOD               : int  2001 2002 2000 2001 2002 2003 2004 2012
2013 2013 ...
 $ Time.period               : logi  NA NA NA NA NA NA ...
 $ OBS_VALUE                 : num  6.1 6.1 6.1 6.1 6.1 6.1 6.1 6.1 6.1 6.1 ...
 $ Observation.value         : logi  NA NA NA NA NA NA ...
 $ DECIMALS                  : int  1 1 1 1 1 1 1 1 1 1 ...
 $ Decimals                  : chr  "One" "One" "One" "One" ...
 $ OBS_STATUS                : chr  "" "" "" "" ...
 $ Observation.status        : chr  "" "" "" "" ...
 $ OBS_STATUS2               : logi  NA NA NA NA NA NA ...
 $ Observation.status.2      : logi  NA NA NA NA NA NA ...
 $ OBS_STATUS3               : logi  NA NA NA NA NA NA ...
 $ Observation.status.3      : logi  NA NA NA NA NA NA ...
```

```
 $ UNIT_MULT                    : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Unit.multiplier              : chr  "Units" "Units" "Units" "Units" ...
```

1.4 Save your data

Now that you have imported your data, save them locally on your computer to use in the second part of the homework. Explain which file format you chose and why you selected this format. [2 pts]

Any rectangular format works. Here, I saved the data in a csv format.

```
# Save health expenditures
write.csv(
  df_WorldBank_health,
  "data/df_WorldBank_health.csv"
  )

# Save life expectancy
write.csv(
  df_WorldBank_life_expectancy,
  "data/df_WorldBank_life_expectancy.csv"
  )
write.csv(
  df_oecd_life_expectancy,
  "data/df_oecd_life_expectancy.csv"
  )

# Save perceived health status
write.csv(
  df_oecd_phs,
  "data/df_oecd_phs.csv"
  )
```

Homework Part 2 (due December 10th, 2025, 23:59)

The relationship between health expenditures and health outcomes has long been a central topic in health economics. Empirical studies show a positive but non-linear relationship between health spending and outcomes such as life expectancy, particularly in low- and middle-income countries.

In high-income settings, however, diminishing returns are evident. For example, the United States spends more per capita on health than any other nation, yet its health outcomes often lag behind those of countries with lower spending (Anderson, Reinhardt, Hussey, & Petrosyan (2003)).

In this group project, we investigate the relationship between health expenditures and health outcomes. The objectives of part 2 are the following:

1. Clean the data
2. Merge the different dataframes
3. Formulate a hypothesis to guide your research
4. Explore the data
5. Visualize the data to validate or invalidate your hypothesis.

2.1 R Setup

```r
library(dplyr)
library(tidyr)
library(ggplot2)

# Optional:
library(skimr)
library(summarytools)
library(forcats)
library(ggrepel)

# Import data
df_WorldBank_health <- read.csv(
  "data/df_WorldBank_health.csv"
  )
df_WorldBank_life_expectancy <- read.csv(
  "data/df_WorldBank_life_expectancy.csv"
  )
df_oecd_life_expectancy <- read.csv(
  "data/df_oecd_life_expectancy.csv"
  )
df_oecd_phs <- read.csv(
  "data/df_oecd_phs.csv"
  )
```

2.2 Data cleaning

2.2.1 Cleaning and summarizing data on healthcare expenditures, life expectancy and perceived health status

Explore the data to identify the relevant variables for your analysis, making sure their formats and values are correct. Since we imported full datasets in Part One, here are some restrictions we want to make on the data we import:

- We are interested in life expectancy at birth for the overall population;
- We will use the following definition for perceived health status: *percentage of the population, aged 15 years old and over who report their health to be 'good/very good' (or excellent)*. This means: we restrict our sample to `AGE` = "aged 15 years and older" and reported health status to "good/very good (or excellent)";
- The World Bank data contains a variable on the economic status of each country. Even though we did not specifically ask to extract this variable in Part One, we will use it in Part Two.

Tasks: For each dataset,

- Perform all the necessary cleaning steps, and clearly describe these steps in the code.
- Comment on the cleaning choices you made regarding the data.
- Keep only the variables of interest in your dataset.
- Summarize your variables of interest.

[2 pts per dataset, 8 pts in total]

Hints:

- Dealing with potential missing values (`NA`s) are an important part of cleaning data. Make sure you document how you deal with them.
- The function `skim` from the package `skimr` as well as the function `summarytools::dfSummary()` can be helpful to get a quick overview of the data. You don't need to print the output in your final solution.
- Remember, our ultimate goal is to study the relationship between healthcare expenditures and healthcare quality by comparing countries. Since you will need to merge multiple datasets later, keep this in mind as you perform your cleaning steps.

2.2.2 Solution

2.2.2.1 Health expenditure from the World Bank

- Remove “aggregates”, i.e. groups of countries.
- The other values are plausible.
- Ignore the longitude and latitudes as well as the capitals. The `lending` variable is also ignored.
- ind_WorldBank_health_gdp: Mean 6.2, min 1.1, max 24.2, 12.5% NAs (617)
- ind_WorldBank_health_usd: Mean 948.9, min 4.2, max 12434.4, 12.5% NAs (619)
- NA’s for specific countries, which we will remove when merging
- income: 4 categories, 23 obs “not classified”, distribution is important
- countries: 4945 countries
- year: 2000 to 2022

```r
# Check
names(df_WorldBank_health)

# dfSummary is commented out because it does not print well in typst/quarto
# dfSummary(df_WorldBank_health)

# Remove aggregates
df_WorldBank_health <- df_WorldBank_health |>
  filter(region != "Aggregates")

# Keep the relevant variables and values
df_WorldBank_health <- df_WorldBank_health |>
      select(iso3c,   year,   income,   country,   ind_WorldBank_health_gdp,
ind_WorldBank_health_usd) |>
  filter(iso3c != "") |>
  rename(
    country = iso3c,
    country_name = country
  )

# Arrange df (optionnal)
df_WorldBank_health <- df_WorldBank_health |>
  arrange(country_name, year)

# Explore NA
df_WorldBank_health |>
  filter(is.na(ind_WorldBank_health_gdp))

# Check distributions with frequency tables
table(df_WorldBank_health$country,
      df_WorldBank_health$income)
```

2.2.2.2 Life expectancy at birth from WDI

- Data almost ready to use.
- 6003 countries (more than in the expenditure variable)
- year: 2000 to 2022
- ind_WorldBank_lifeexpectancy:
  - Mean (sd) : 70.4 (8.6)
  - min < med < max: 14.7 < 71.9 < 86.2
  - NAs: 0

```
# Checks
# dfSummary(df_WorldBank_life_expectancy)
names(df_WorldBank_life_expectancy)

# Keep the relevant variables and values
df_WorldBank_life_expectancy <- df_WorldBank_life_expectancy |>
  select(iso3c, year, SP.DYN.LE00.IN) |>
  filter(iso3c != "") |>
  rename(
    country = iso3c,
    ind_WorldBank_lifeexpectancy = `SP.DYN.LE00.IN`
  )
```

2.2.2.3 Life expectancy from OECD

This dataset needs more work.

- We need to filter out gender-specific observation (gender = "F" or "M").
- We see that many variables have no variation: AGE is always Y0 because we have life expectancy at birth.
- We will keep only ObsValue, MEASURE, REF_AREA, TIME_PERIOD.
- Because we merge the data together, we change the variable names.
- Since we will merge data together, we restrict to the years after 2000.
- 1170 countries
- life_expectancy:
  - Mean (sd) : 77.3 (4.9)
  - min < med < max: 53.9 < 78.4 < 84.6
  - No NA's
  - Higher than world bank. Expected because OECD countries.

```
# Check the names of the variables
# dfSummary(df_oecd_life_expectancy)
names(df_oecd_life_expectancy)

# Filter for life expectancy at birth for overall pop (all men and women)
df_oecd_life_expectancy <- df_oecd_life_expectancy |>
  filter(SEX == "_T")
```

```
# We only keep the relevant variables for our analysis. We rename them for
# future use.
df_oecd_life_expectancy <- df_oecd_life_expectancy |>
  select(REF_AREA, TIME_PERIOD, ObsValue, MEASURE) |>
  rename(
    country = REF_AREA,
    year = TIME_PERIOD,
    life_expectancy_oecd = ObsValue
  )

# Only keep observations after 2000. We restricted to 2000-2022 in Part One.
df_oecd_life_expectancy <- df_oecd_life_expectancy |>
  filter(
    year >= 2000
  )

# Check countries
count(df_oecd_life_expectancy, country)
```

2.2.2.4 Perceived Health status from OECD

Steps:

- Filter out gender-specific observation (gender = “F” or “M”). Following our definition:
- We keep age = “aged 15 years and older”.
- We keep the variables for those who report their health to be ‘good/very good’ (or excellent) (`HEALTH_STATUS`)
- Keep only `SOCIO_ECON_STATUS = '_z'`.
- Students must also be aware of the different definitions in the `OBS_STATUS` variable. For simplification, we keep all the observations, but it needs to be noted.
- We restrict to years >= 2000.
- 666 countries
- health_status_oecd:
  - Mean (sd) : 68.4 (12.2)
  - min < med < max: 30 < 69.9 < 91.
  - No NAs

```
# Check names
names(df_oecd_phs)

# dfSummary commented out for clean rendering
# dfSummary(df_oecd_phs)

# Filter for perceived health status
df_oecd_phs <- df_oecd_phs |>
  filter(SEX == "_T") |> # all genders
  filter(AGE == "Y_GE15") |>   # aged 15 years old and over
```

```
  filter(HEALTH_STATUS == "G") # who report their health to be ‘good/very
good' (or excellent)

# We also focus on non-defined Socio-econ status
df_oecd_phs <- df_oecd_phs |>
  filter(SOCIO_ECON_STATUS == "_Z")

## We have more than one observations per country-year, because of
Observation.status
# B: Time-series break
# D: definition differs.
# We'll keep all of them for now
count(df_oecd_phs, OBS_STATUS)

# Keep the relevant variables and values
df_oecd_phs <- df_oecd_phs |>
  select(REF_AREA, AGE, TIME_PERIOD, ObsValue, OBS_STATUS) |>
  rename(
    country = REF_AREA,
    age = AGE,
    year = TIME_PERIOD,
    health_status_oecd = ObsValue
  )

# Restrict date
df_oecd_phs <- df_oecd_phs |>
  filter(
    year >= 2000
  )

# Check countries
# Not the same amount of observations per country
count(df_oecd_phs, country)
```

2.3 Measuring healthcare quality

2.3.1 Proxies for quality

We study the relationship between healthcare expenditures and healthcare quality. For healthcare quality, we are using life expectancy and perceived health status. In statistical jargon, we are using these two variables as *proxies* of healthcare quality. Do a quick research and write down a definition of **proxy variables** that you understand and will remember. Give a short example. Explain briefly how our two measures might be good or bad proxies for healthcare quality. [2 pts]

2.3.1.1 Solution

Proxy variables are measurable variables that are used to represent or substitute for another variable that is difficult or impossible to measure directly. They serve as indirect indicators of the concept of interest.

Life expectancy as a proxy for healthcare quality:

- **Good proxy**: Reflects overall population health outcomes and healthcare system effectiveness in preventing premature death.
- **Limitations**: Influenced by factors beyond healthcare (socioeconomic conditions, lifestyle, environment, genetics). Health shocks can influence life expectancy (look at the Covid years in the data, or at the life expectancy for Ukraine in 2021-2023.)

Perceived health status as a proxy for healthcare quality:

- **Good proxy**: Captures patient experience and subjective wellbeing, which are important healthcare outcomes.
- **Limitations**: Subjective measure that varies across cultures and individuals; may not reflect objective medical conditions, or objective treatment success.

2.3.2 Choosing between OECD and World Bank for life expectancy

In practice, it is common to have different sources for the same indicator. Would you expect life expectancy to differ between the two datasets (i.e., the one from World Bank and the one from OECD)? Why? Compare both indicators and show their similarities and/or dissimilarities. Which one should we pick? [1 pt]

2.3.2.1 Solution

- Any test is good.
- I merged the data using **left join**, and see if there were discrepancies in the indicators testing for their ratios (if ratio = 1, they are

exactly the same). Many unmerged observations since OECD data cover only OECD countries. The indicators are very similar, so I would keep only the World Bank for now.

In principle, for a given year and country, both datasets should report the same life expectancy value. However, there might be differences due to differences in country or year coverage, data collection methods, or rounding practices.

```
merged <- left_join(df_WorldBank_life_expectancy,
                    df_oecd_life_expectancy,
                    by = c("country", "year")) |>
  mutate(test = ind_WorldBank_lifeexpectancy/life_expectancy_oecd)

summary(merged$test)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
  0.930   0.998   0.999   0.998   1.000   1.022    4833 
```

2.4 Merge the data

Merge the datasets in a single dataset that contains all the information that will be used for the subsequent analysis. Answer the following questions in text:

1. Which variable(s) will you use to perform the merge? [1pt]
2. Are there any additional cleaning steps required before merging the datasets? [1pt]
3. What type of merge is appropriate (e.g., right join, left join, full join, etc.)? [1pt]
4. Are there any steps necessary in cleaning data after the merge? Pay attention to the missing values (if any) and explain how you deal with them. [1pt]

2.4.1 Solution

1. Merge keys: We use `country` and `year` as the key variables to match observations across all datasets.
2. Pre-merge cleaning: Before merging, ensure that country codes and year formats are consistent across datasets. Time periods should overlap between OECD and WorldBank data. We already checked for this in the previous steps.
3. Merge type: I use `full_join()` for all three merges. This preserves all observations from both datasets and creates NAs where country-year combinations exist in only one dataset, allowing us to identify unmatched observations. My intention is to use as many data points

as possible in the analysis, making sure that I have good coverage in my time series. Other solutions with `left_join()` or an `inner_join()` are also possible, and have different consequences. For instance, `inner_join()` restricts the analysis to OECD countries. This is also correct, but needs to be commented.

4. Post-merge cleaning:
    - First, merge within data sources (OECD datasets together, World-Bank datasets together)
    - Then merge the combined datasets, using suffixes `"_wb"` and `"_oecd"` to distinguish overlapping variable names
    - Remove redundant columns: `MEASURE`, `age`, `OBS_STATUS`
    - Arrange by country and year for easier inspection
    - Remove countries where health expenditure is missing for more than half the observations (> 50% NAs), as they provide insufficient data for time series analysis. This is a consequence of the `inner_join()`, as the join created NAs in the expenditure indicator. Since I am working with a time series, I am OK with having some NAs remaining in my dataset. These will be taken care of separately for each analysis (in the `na.rm = T` arguments of my functions).

```
# Merge OECD data
df_oecd <- df_oecd_life_expectancy |>
  full_join(
    df_oecd_phs, by = c("country", "year")
  )

# Merge WorldBank data
df_world_bank <- df_WorldBank_life_expectancy |>
  full_join(
    df_WorldBank_health, by = c("country", "year")
  )

# Merge the two datasets
df_final <- df_world_bank|>
  full_join(
    df_oecd,
    by = c("country", "year"),
    suffix = c("_wb", "_oecd")
  )

df_final <- df_final |>
  select(-MEASURE, -age, -OBS_STATUS)
```

```
# Checks
# dfSummary(df_final)

# Arrange
df_final <- df_final |>
```

```r
  arrange(country, year)

# Remove countries with number of  NAs > 0.5 in the Worldbank indicator
set_countries_NA <- df_final |>
  group_by(country) |>
  summarise(N = n(),
            NA_exp = sum(is.na(ind_WorldBank_health_usd))) |>
  filter(NA_exp > N/2)

df_final <- df_final |>
  filter(!country %in% set_countries_NA$country)
```

2.5 Explore and visualize the data

Hints/Remarks: In the following questions, there are different ways of giving a correct answer. As seen in the course, you might choose to present numbers in text, tables, or plots, or a mixture of these different methods. Choose the method you find the most effective to convey your answer. Remember to be concise.

2.5.1 Trends in expenditures

Let's start exploring the healthcare expenditure variable. How did general healthcare expenditure in USD evolve over time? Is this trend similar in healthcare expenditures as a share of gdp? What should we be aware of/careful when interpreting the trend? [2pts]

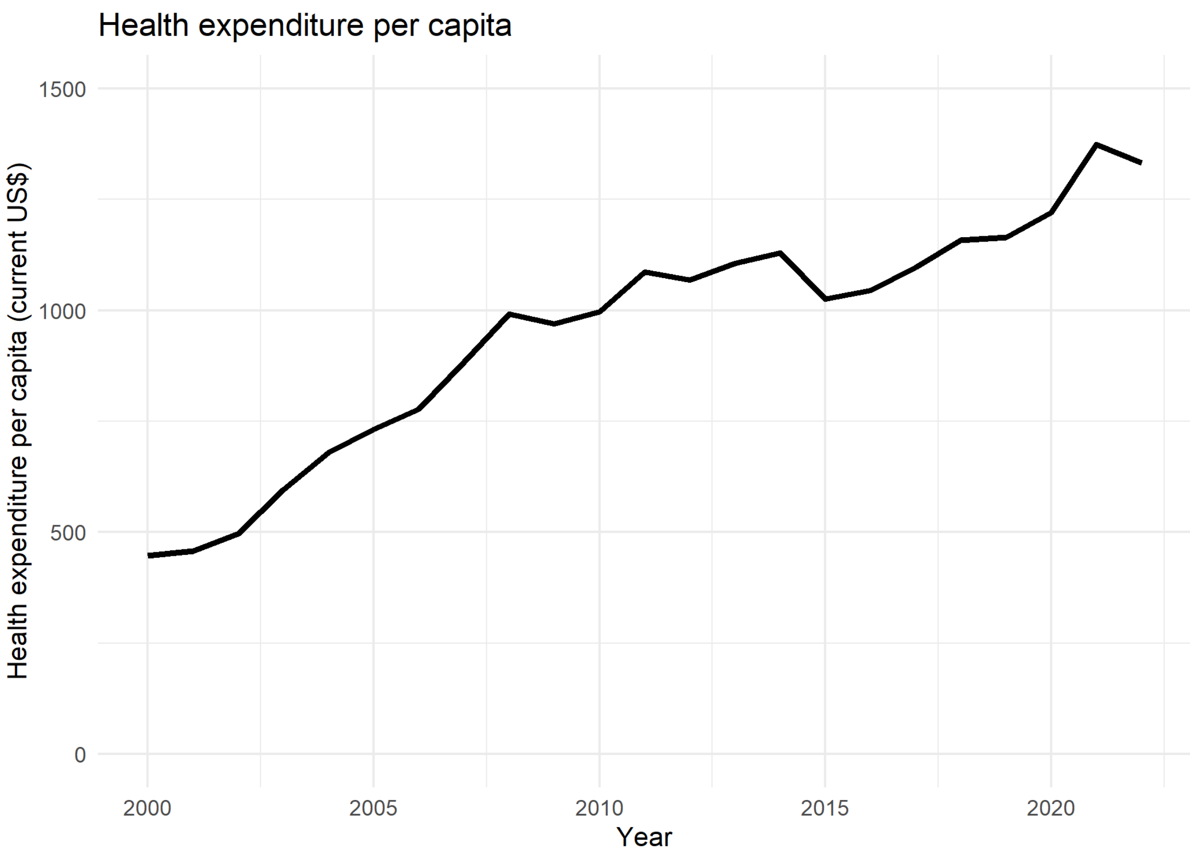
Positive trend.

Careful:

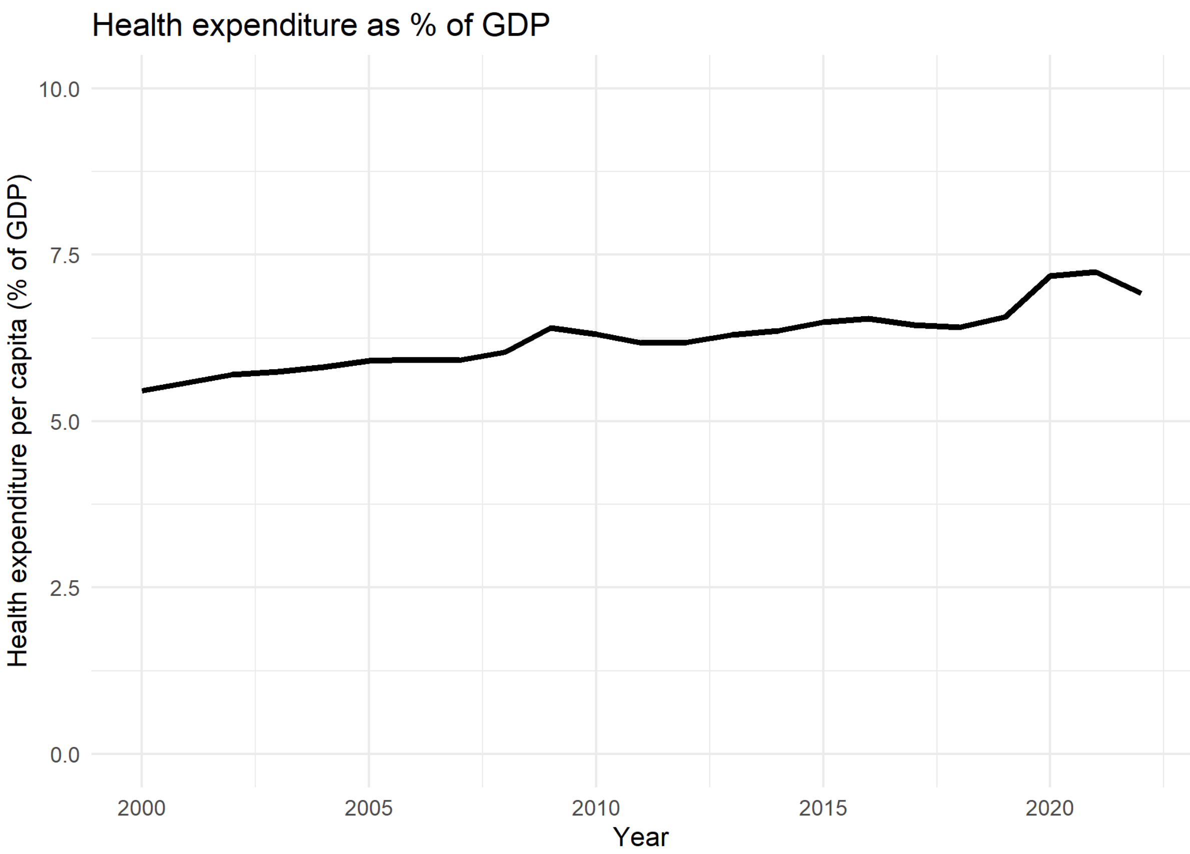
- Indicators in current USD are not adjusted for inflation or purchasing power… This makes comparisons more difficult, especially for low-income countries.
- For share of gdp: suitable for comparing across countries (within a year) or over time (roughly), though the latter can still be influenced by GDP volatility or currency shifts.

2.5.1.1.1 Solutions

```
df_final |>
  group_by(year) |>
  summarise(ind_WorldBank_health_usd = mean(ind_WorldBank_health_usd, na.rm
= T)) |> # Take care of missing values
  ggplot(aes(x = year, y = ind_WorldBank_health_usd)) +
  geom_line(linewidth=1.2) +
  labs(
    title = "Health expenditure per capita",
    x = "Year",
    y = "Health expenditure per capita (current US$)"
  ) +
  scale_y_continuous(limits = c(0, 1500)) +
  theme_minimal()
```




```
df_final |>
  group_by(year) |>
  summarise(ind_WorldBank_health_gdp = mean(ind_WorldBank_health_gdp, na.rm
= T)) |> # Take care of missing values
  ggplot(aes(x = year, y = ind_WorldBank_health_gdp)) +
  geom_line(linewidth=1.2) +
  labs(
    title = "Health expenditure as % of GDP",
    x = "Year",
    y = "Health expenditure per capita (% of GDP)"
  ) +
  scale_y_continuous(limits = c(0, 10)) +
  theme_minimal()
```



2.5.2 Expenditures per income group

We suspect the expenditure trends between countries and between years to be large, especially between the different income groups.

Explore the trends per income group. Make sure the income groups are correctly ordered and coded. Which income group shows the steepest increase in per-capita health spending over the last decades? [2pts]

2.5.2.1 Solutions

- Recode to factors with correct levels
- Create means per groups. We do not weight countries, each country counts as one.

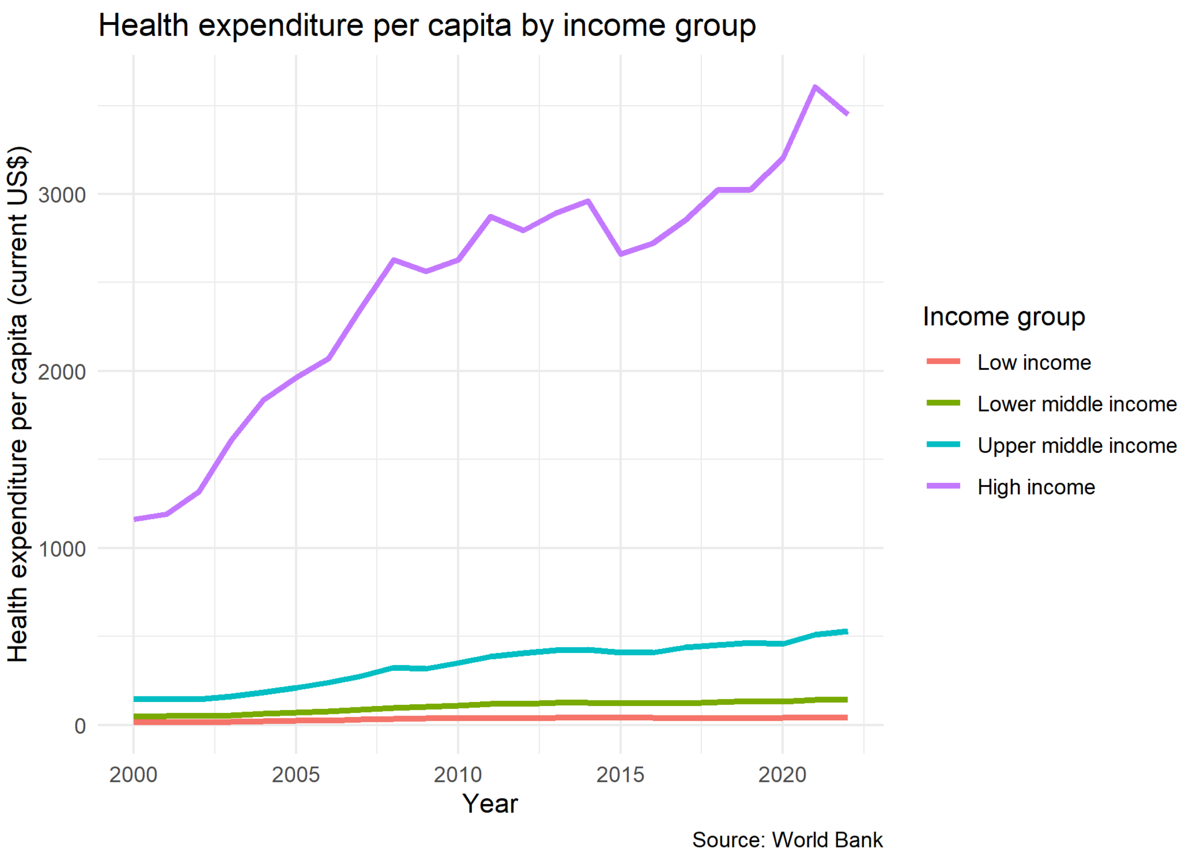
```
# WDI: healthcare costs
count(df_final, income)
```

```
               income    n
1         High income 1426
2          Low income  529
3 Lower middle income 1173
4 Upper middle income 1219
```

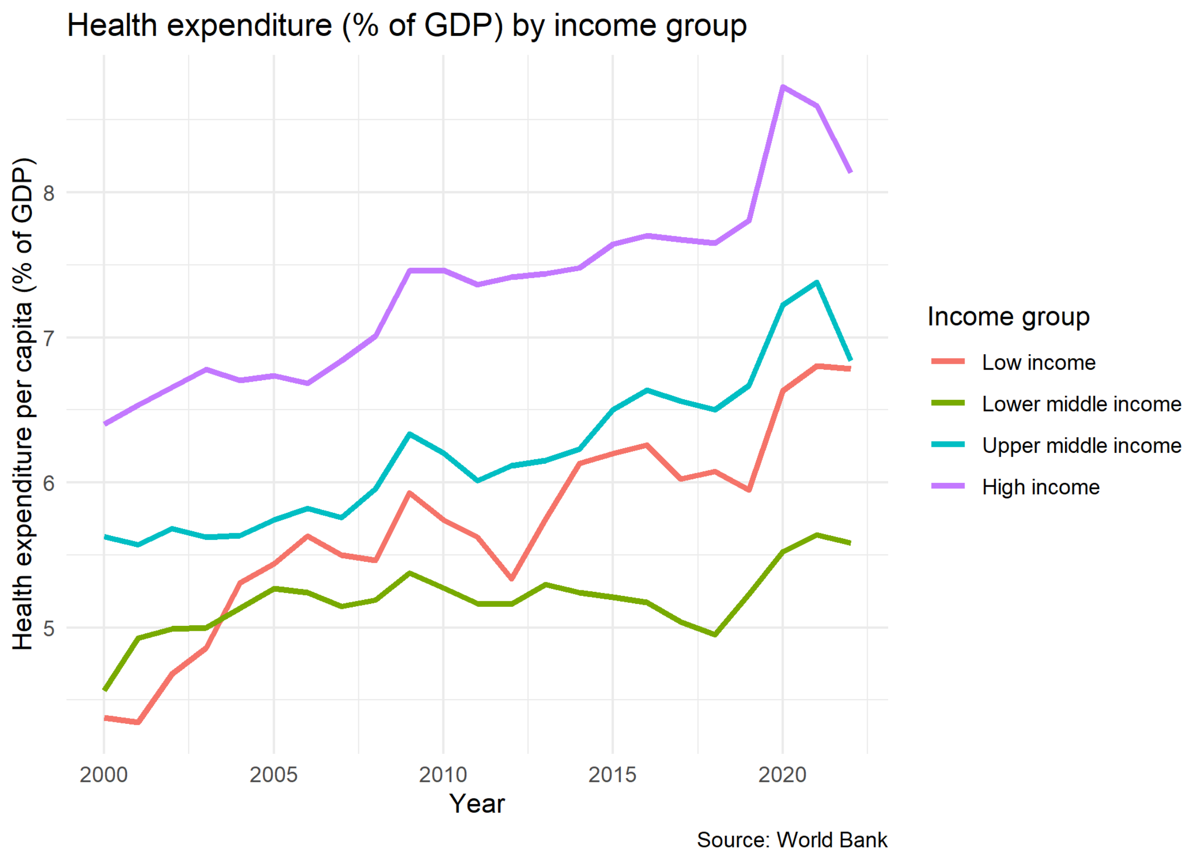
```r
typeof(df_final$income)
```

```
[1] "character"
```

```r
income_groups <- c("Low income", "Lower middle income", "Upper middle income",
"High income")

df_final <- df_final |>
    mutate(income = factor(income, levels = income_groups))

# Create data
data_income_groups <- df_final |>
  filter(income %in% income_groups) |>
  group_by(year, income) |>
  summarise(
     ind_WorldBank_health_gdp_mean = mean(ind_WorldBank_health_gdp, na.rm =
T),
    ind_WorldBank_health_gdp_sd = sd(ind_WorldBank_health_gdp, na.rm = T),
    ind_WorldBank_health_gdp_coeffvar = sd(ind_WorldBank_health_gdp, na.rm =
T)/mean(ind_WorldBank_health_gdp, na.rm = T),
     ind_WorldBank_health_usd_mean = mean(ind_WorldBank_health_usd, na.rm =
T),
    ind_WorldBank_health_usd_sd = sd(ind_WorldBank_health_usd, na.rm = T),
    ind_WorldBank_health_usd_coeffvar = sd(ind_WorldBank_health_usd, na.rm =
T)/mean(ind_WorldBank_health_usd, na.rm = T)
    )

# Plot expenditure by income group
data_income_groups |>
  ggplot(aes(x = year, y = ind_WorldBank_health_usd_mean, color = income)) +
  geom_line(linewidth=1.2) +
  labs(
    title = "Health expenditure per capita by income group",
    x = "Year",
    y = "Health expenditure per capita (current US$)",
    color = "Income group",
    caption = "Source: World Bank"
  ) +
  theme_minimal()
```




```r
# Plot expenditure by income group
data_income_groups |>
  ggplot(aes(x = year, y = ind_WorldBank_health_gdp_mean, color = income)) +
  geom_line(linewidth=1.2) +
  labs(
    title = "Health expenditure (% of GDP) by income group",
    x = "Year",
    y = "Health expenditure per capita (% of GDP)",
    color = "Income group",
    caption = "Source: World Bank"
  ) +
  theme_minimal()
```



We might also want to look at the spread (distribution of variables and variance). This was not explicitly asked, so not part of grading.

```
df_final |>
  ggplot(aes(x = income, y = ind_WorldBank_health_usd)) +
  geom_boxplot()

df_final |>
  ggplot(aes(x = income, y = ind_WorldBank_health_gdp)) +
  geom_boxplot()
```

2.5.3 Life expectancy

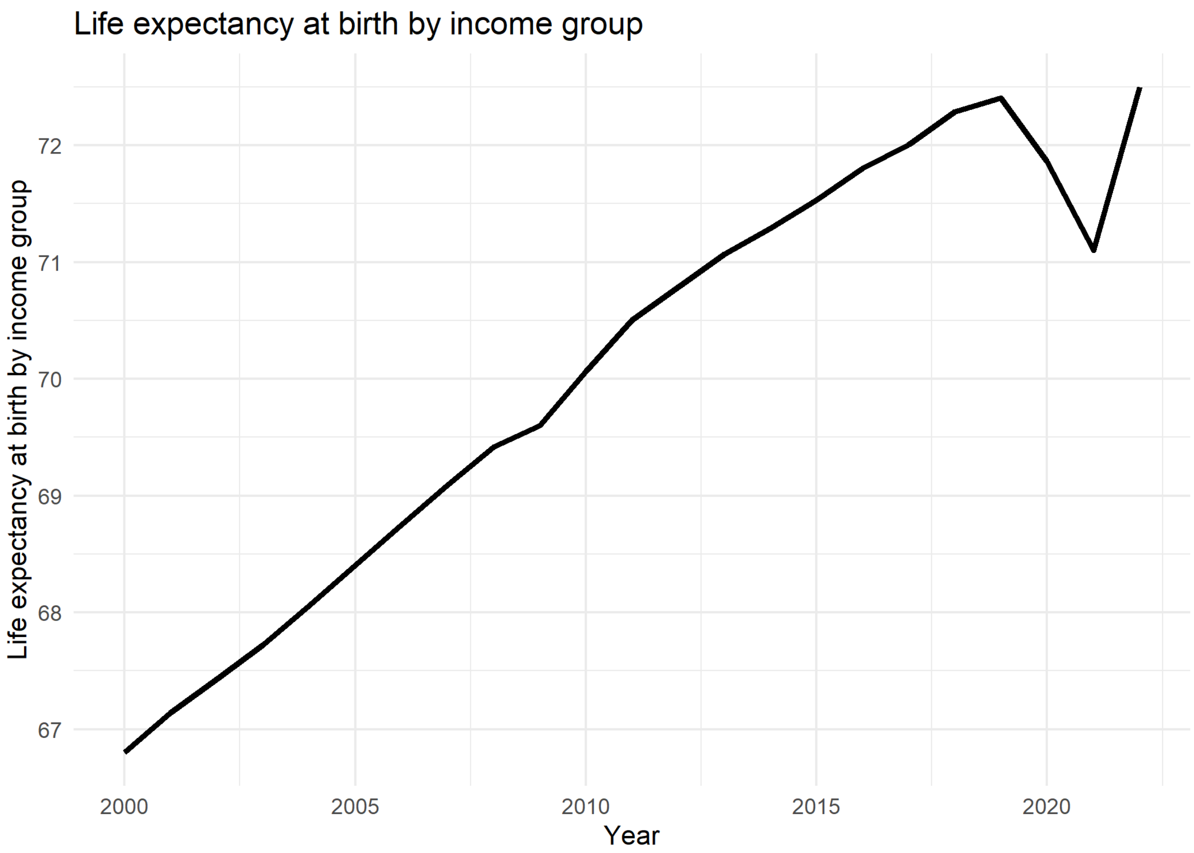
Look at the trends in life expectancy by country income group using the WDI data. Briefly comment. Do life-expectancy curves for all groups rise at a similar pace, or do some groups converge or diverge over time? [2pts]

2.5.4 Solutions

All groups exhibit a long-term upward trajectory, indicating rising life expectancy (or spending) over two decades. Gaps between high- and low-income groups persist, though lower-income groups show faster relative growth.

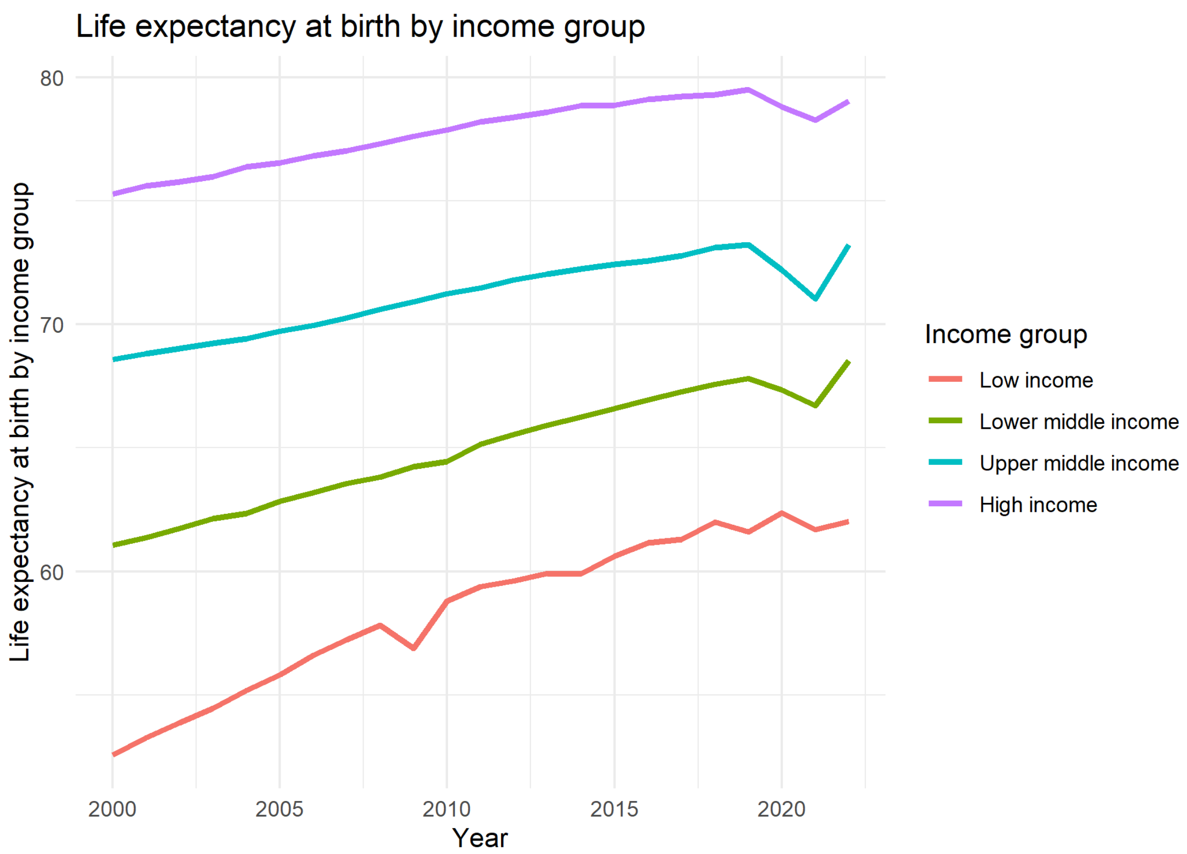
```
Around 2020-2021 there's a pronounced dip for every group, reflecting
the the COVID-19 pandemic, with most groups rebounding in 2022.
```

```
df_final |>
  group_by(year) |>
  summarize(
    ind_WorldBank_lifeexpectancy = mean(ind_WorldBank_lifeexpectancy, na.rm
= T)
  ) |>
  ggplot(aes(x = year, y = ind_WorldBank_lifeexpectancy)) +
  geom_line(linewidth=1.2) +
  labs(
    title = "Life expectancy at birth by income group",
    x = "Year",
    y = "Life expectancy at birth by income group",
    color = "Income group"
  ) +
  theme_minimal()
```




```
df_final |>
  filter(income %in% income_groups) |>
  group_by(year, income) |>
  summarize(
    ind_WorldBank_lifeexpectancy = mean(ind_WorldBank_lifeexpectancy, na.rm
= T)
```

```r
) |>
ggplot(aes(x = year, y = ind_WorldBank_lifeexpectancy, color = income)) +
geom_line(linewidth=1.2) +
labs(
  title = "Life expectancy at birth by income group",
  x = "Year",
  y = "Life expectancy at birth by income group",
  color = "Income group"
) +
theme_minimal()
```



2.5.5 Perceived health status

Let's look at our perceived health status indicator. Provide descriptive statistics of health status over time and per country. You are free to present these statistics in different ways. Choose the option that best fits your goal. [2pts]

2.5.5.1 Solutions

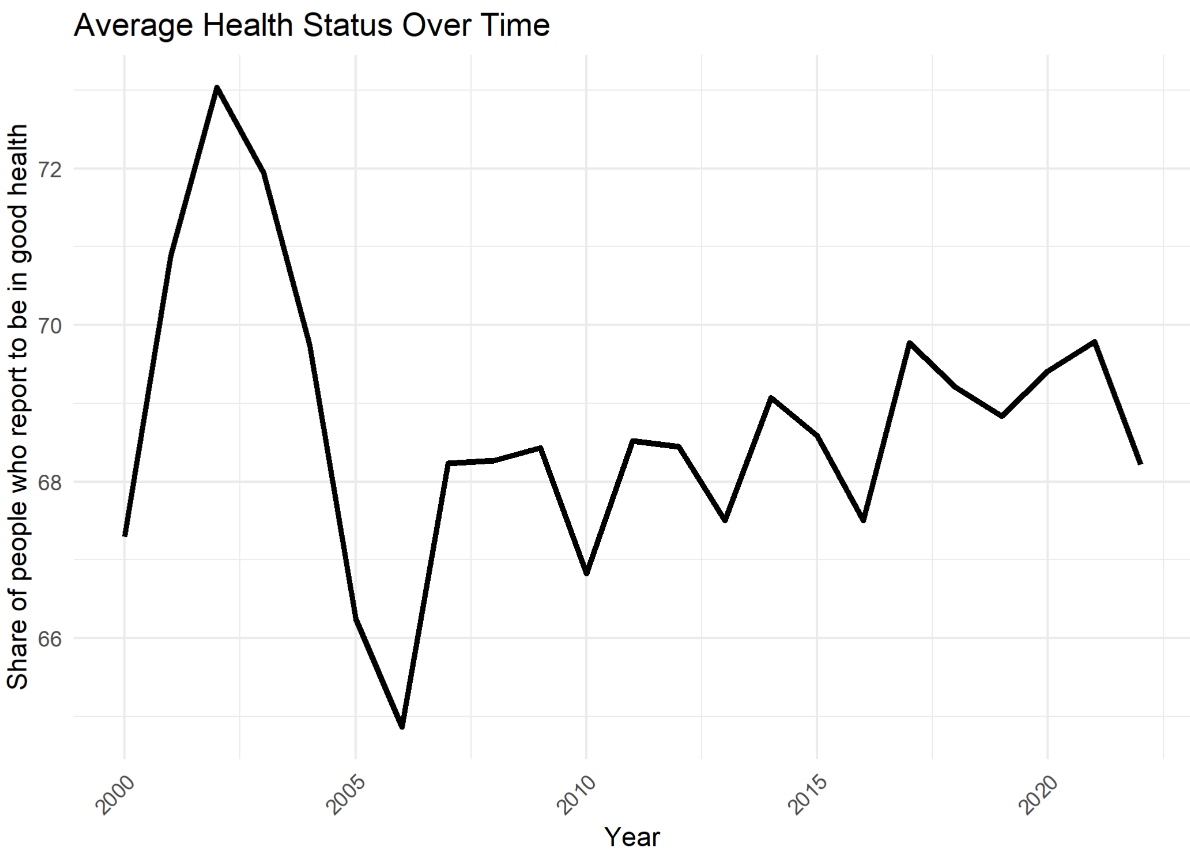
Data before 2005 seem to be unreliable. Also, only OECD countries, so only higher income countries. Only two countries in middle-high income.

The health status of the individuals is mostly stable over time, with a spike in the early 2000 for each group.

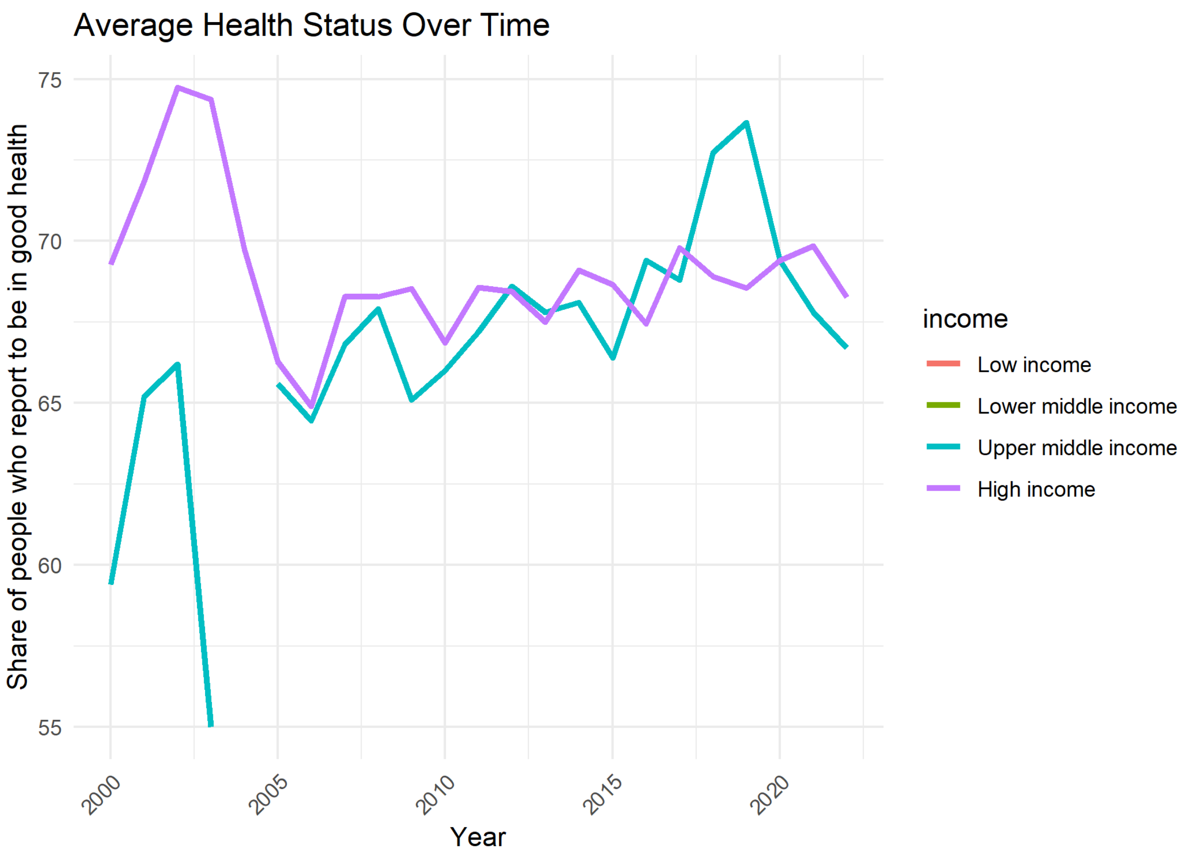
```
# Summary
summary(df_final$health_status_oecd)
```

Min. 1st Qu. Median Mean 3rd Qu. Max. NA's 30.0 60.6 69.9 68.4 76.6 91.4 3681

```
# 2, Health status
df_final |>
  group_by(year) |>
  summarise(
    avg_health_status = mean(health_status_oecd, na.rm=TRUE)
  ) |>
  ggplot(aes(x = year, y = avg_health_status)) +
  geom_line(size=1.2) +
  labs(
    title = "Average Health Status Over Time ",
    x = "Year",
    y = "Share of people who report to be in good health"
  ) +
  theme_minimal() +
  theme(
    legend.position = "right",
    axis.text.x = element_text(angle=45, hjust=1)
  )
```




```r
# Health status per income group
df_final |>
  group_by(year, income) |>
  summarise(
    avg_health_status = mean(health_status_oecd, na.rm=TRUE)
  ) |>
  ggplot(aes(x = year, y = avg_health_status, color = income)) +
  geom_line(size=1.2) +
  labs(
    title = "Average Health Status Over Time ",
    x = "Year",
    y = "Share of people who report to be in good health"
  ) +
  theme_minimal() +
  theme(
    legend.position = "right",
    axis.text.x = element_text(angle=45, hjust=1)
  )
```


Average Health Status Over Time
Share of people who report to be in good health
75
70
65
60
55
2000
2005
2010
2015
2020
Year
income
Low income
Lower middle income
Upper middle income
High income

2.6 Three Hypotheses

2.6.1 Formulate your hypotheses

As introduced at the beginning, we're interested in the correlation between healthcare expenditure and healthcare quality at the country level. More specifically, we want to test three hypotheses. To guide you in the formulation of these three hypotheses, answer the following questions without using the data, be as concise as possible. [3pts]

1. Ex ante, would you expect a correlation between healthcare expenditure and healthcare quality? If so, would it be a positive or negative correlation?
2. Do you think health status and life expectancy are correlated? If so, would it be a positive or negative correlation?
3. Do you expect the correlation between expenditures and healthcare quality to be the same for high and low income countries?

2.6.1.1 Solution

1. Yes – likely positive, as more spending is expected to improve healthcare access, infrastructure, and outcomes.
2. Yes – likely positive, since better perceived health is typically associated with longer life.
3. Yes and No – we expect different correlations strengths across income groups, but a positive correlation everywhere. High-income countries may show weaker correlations due to diminishing returns (they are already healthy enough), while low-income countries may show stronger correlations as basic health infrastructure improvements yield larger gains.

2.6.2 Validate (or invalidate) your three hypotheses

Using our data, we want to validate or invalidate our hypotheses empirically.

Hints:

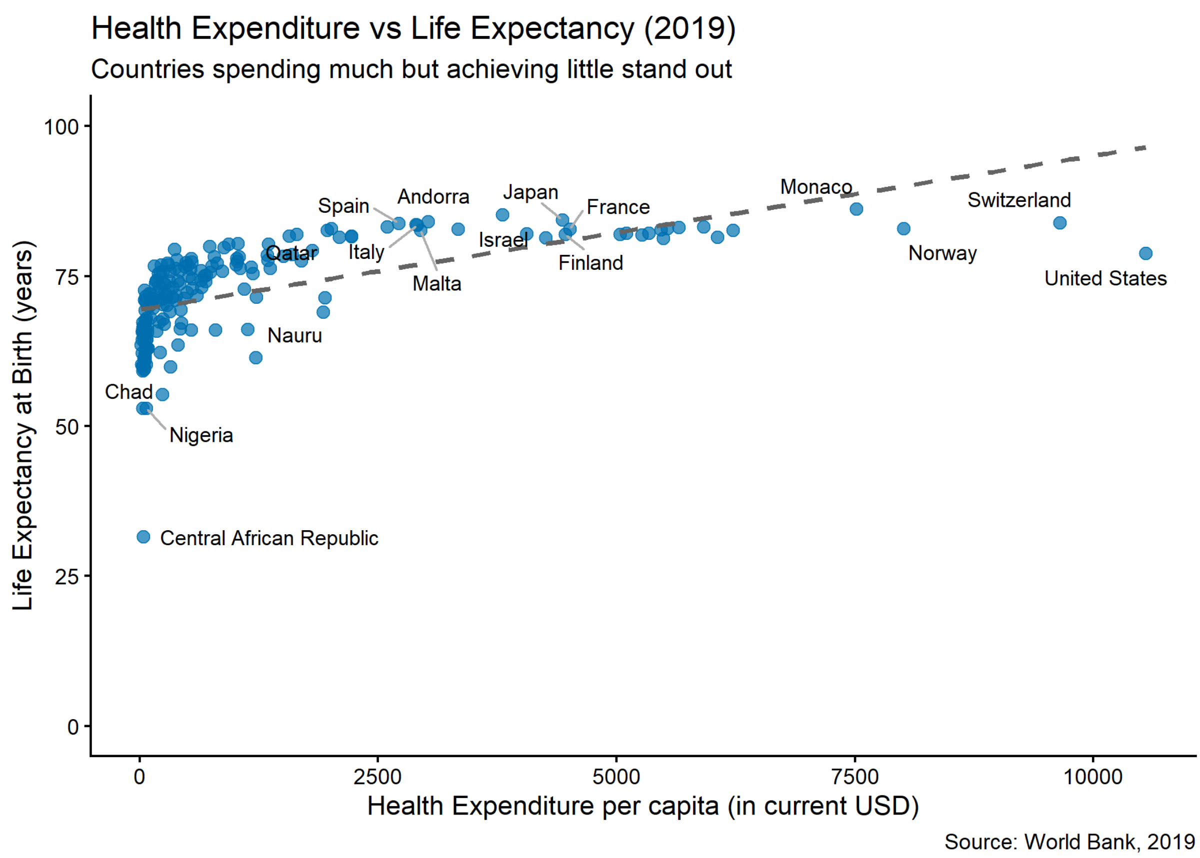
- There are different ways of exploring correlations. We recommend you to create a scatter plot (or multiple) to understand the relationship between health expenditure and life expectancy/perceived health status. The goal of the Data Handling course is to explore the pipeline presented in lecture 1. We do not expect you to conduct inferential statistics for now. You might want to apply the knowledge from your statistics and econometrics courses, but this is not necessary.

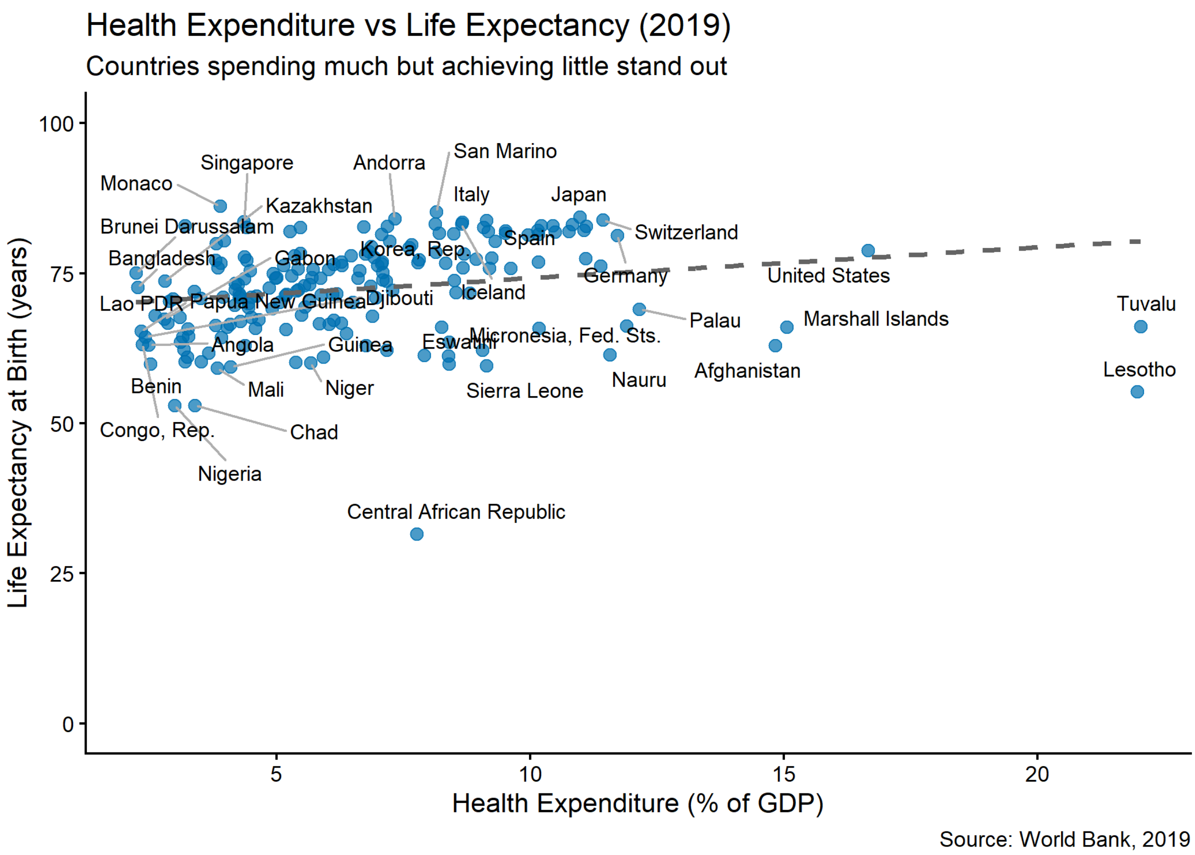
- For simplification, pick the year 2019 for your analyses (before Covid).
- If you present graphs, make sure your graphs follow the recommendations seen in the course.

2.6.2.1 Correlation between healthcare expenditure and healthcare quality.

Explore the correlation with your data. Do your findings confirm your hypothesis? Give a final statement about your results. [3pts]

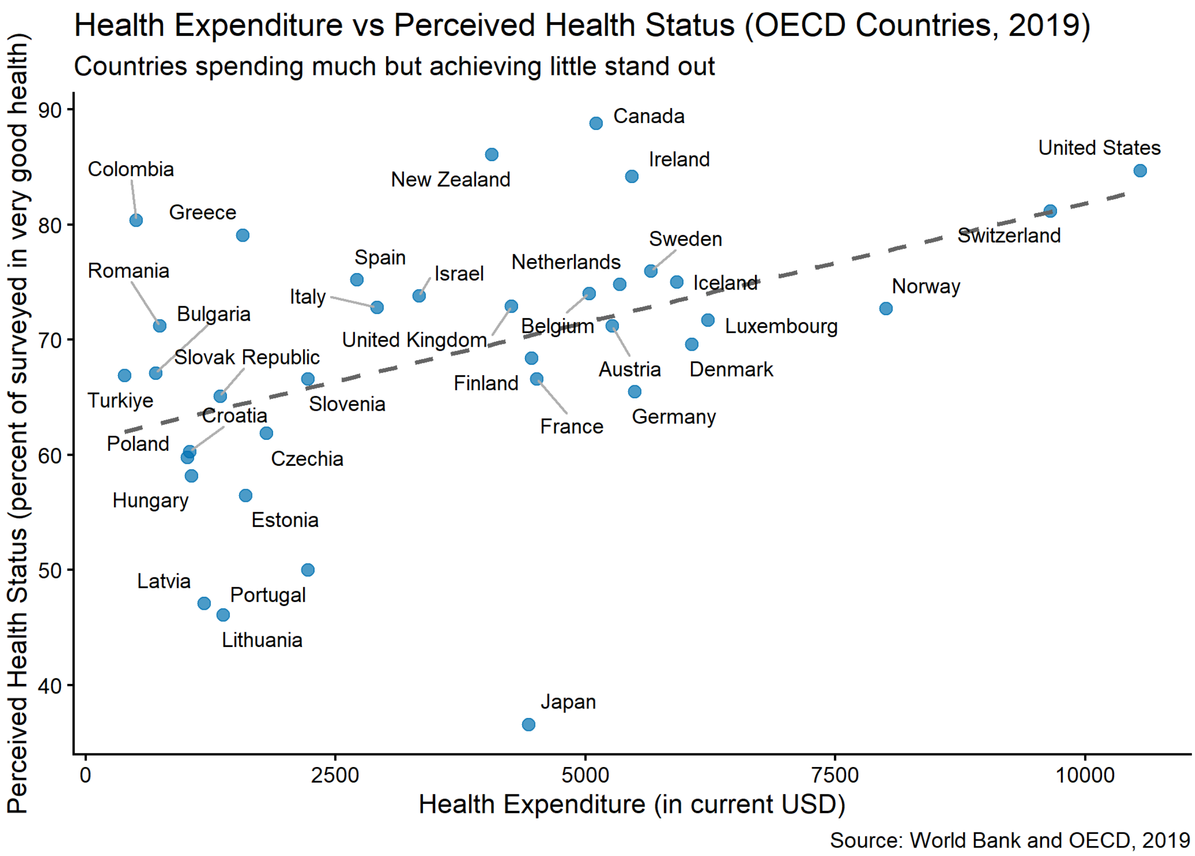
2.6.2.1.1 Solution: life expectancy

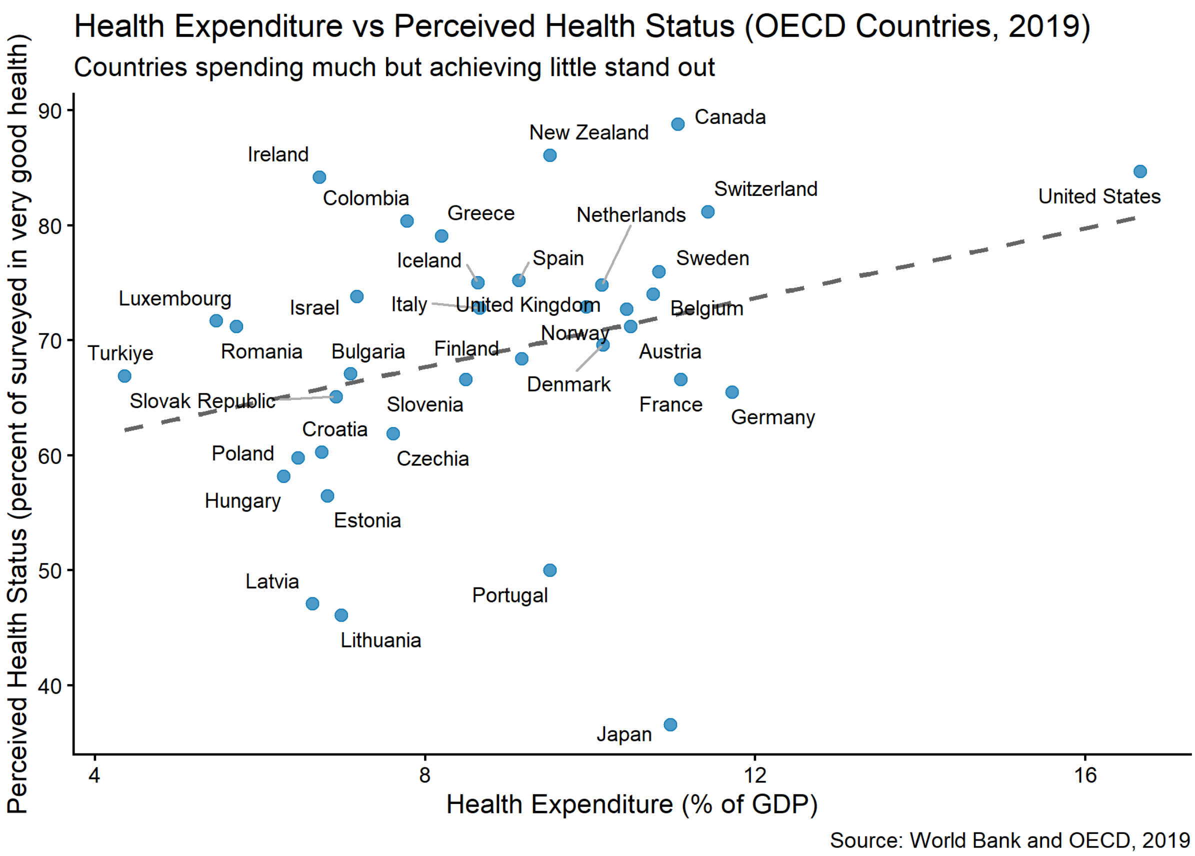




Health Expenditure vs Life Expectancy (2019)
Countries spending much but achieving little stand out
Life Expectancy at Birth (years)
Health Expenditure (% of GDP)
100
75
50
25
0
5
10
15
20
Monaco
Singapore
Andorra
San Marino
Italy
Japan
Kazakhstan
Brunei Darussalam
Switzerland
Spain
Korea, Rep.
Bangladesh
Gabon
Germany
United States
Iceland
Lao PDR
Papua New Guinea
Djibouti
Tuvalu
Palau
Marshall Islands
Micronesia, Fed. Sts.
Eswatini
Angola
Guinea
Afghanistan
Lesotho
Nauru
Benin
Mali
Niger
Sierra Leone
Congo, Rep.
Chad
Nigeria
Central African Republic
Source: World Bank, 2019

2.6.2.1.2 Perceived health status





2.6.2.2 Correlation between healthcare expenditure and healthcare quality: a more precise answer

Let's be more precise about the correlation between the two variables. First, define the measure of correlation that you use. Use the R command `cor` or any other command and then comment about the result. Are they coherent with what your hypothesis? Give a brief economic interpretation to the results. [2pts]

```
# Overall correlation across all income groups
overall_correlations_2019 <- df_final |>
  filter(year %in% c(2005, 2019),
         income %in% income_groups) |>
  group_by(year) |>
  summarise(
    overall_cor_usd = cor(ind_WorldBank_health_usd,
ind_WorldBank_lifeexpectancy, use = "complete.obs"),
    overall_cor_gdp = cor(ind_WorldBank_health_gdp,
ind_WorldBank_lifeexpectancy, use = "complete.obs")
  )

cat("Overall correlations (2019):")
```

```
Overall correlations (2019):
```

```
overall_correlations_2019
```

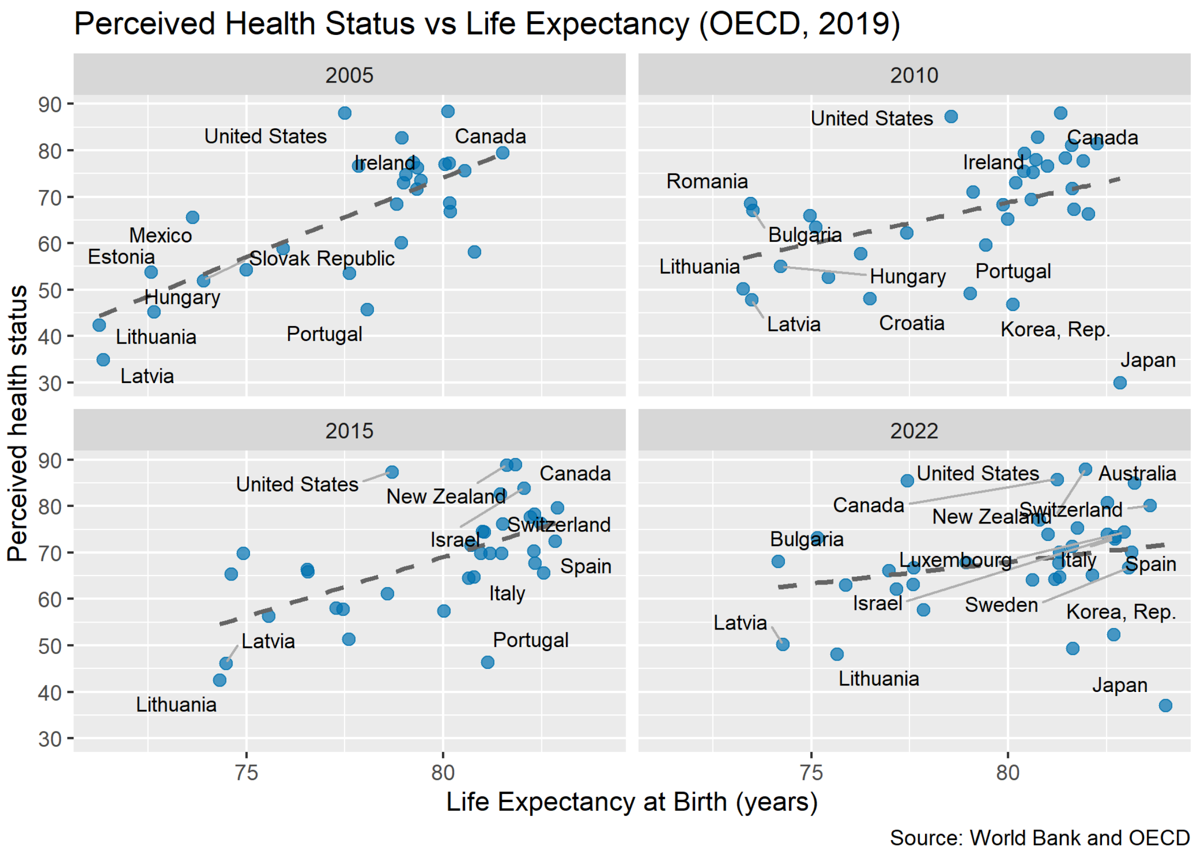
```
# A tibble: 2 × 3
   year overall_cor_usd overall_cor_gdp
  <int>           <dbl>           <dbl>
1  2005           0.569           0.223
2  2019           0.603           0.204
```

```
# Possible with linear regression
lm_usd <- lm(ind_WorldBank_lifeexpectancy ~ ind_WorldBank_health_usd +
                     factor(year) + income,
                     data = df_final)
summary(lm_usd)

lm_gdp <- lm(ind_WorldBank_lifeexpectancy ~ ind_WorldBank_health_gdp +
                     factor(year) + income,
                     data = df_final)
summary(lm_gdp)
```

2.6.2.3 Correlation between health status and life expectancy

Explore the correlation with your data. Do your findings confirm your hypothesis? Give a final statement about your results. [3pts]

- The correlation can be shown only for OECD countries.
- Showing the correlation in 2019 was enough. Out of curiosity, I chose to explore the correlation over time using a facet.
- From this facet graph, it is clear that life expectancy go up over time.
- The correlation is positive, but flattens over the years. It looks like the gap between countries becomes smaller over time.



2.6.2.4 Correlation between expenditures and healthcare quality for high and low income countries?

Explore the correlations by income group to quantify the differences. What do you find. Make a final statement. [3pts]

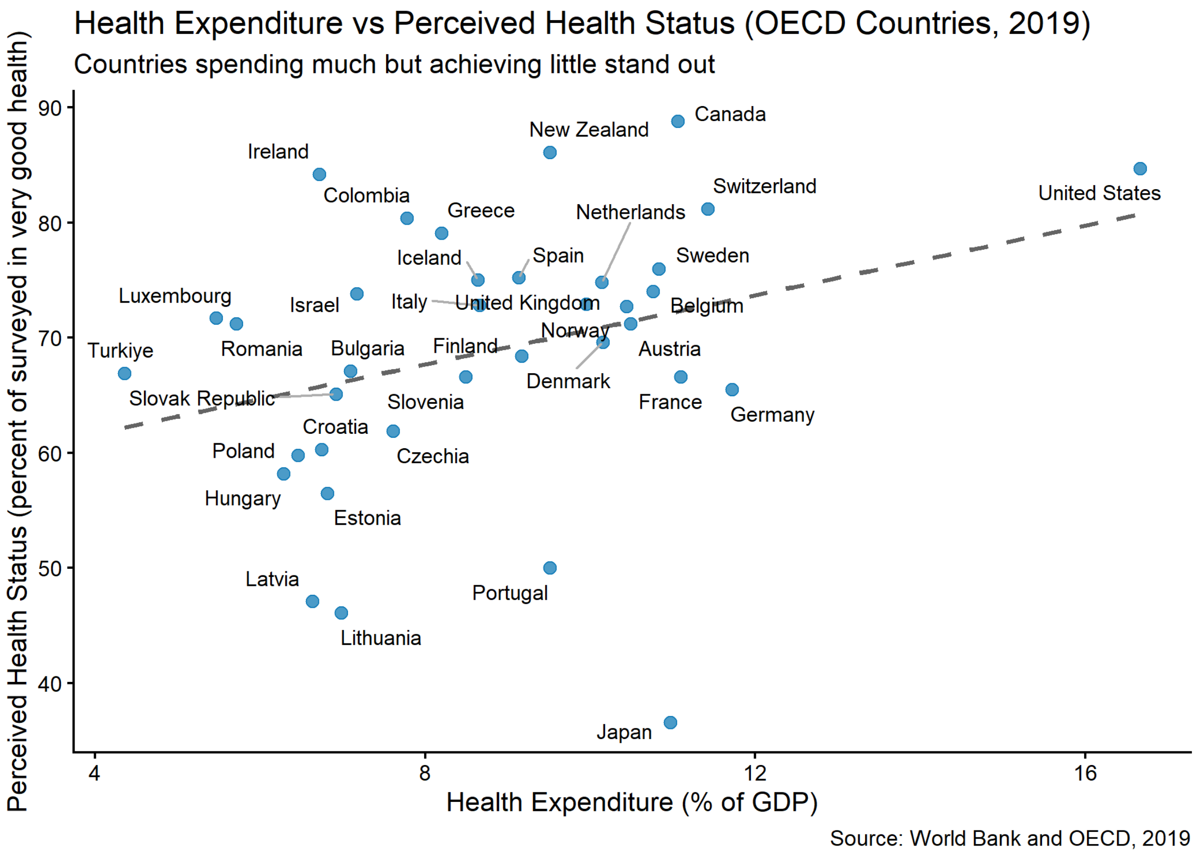
Perceived Health Status:

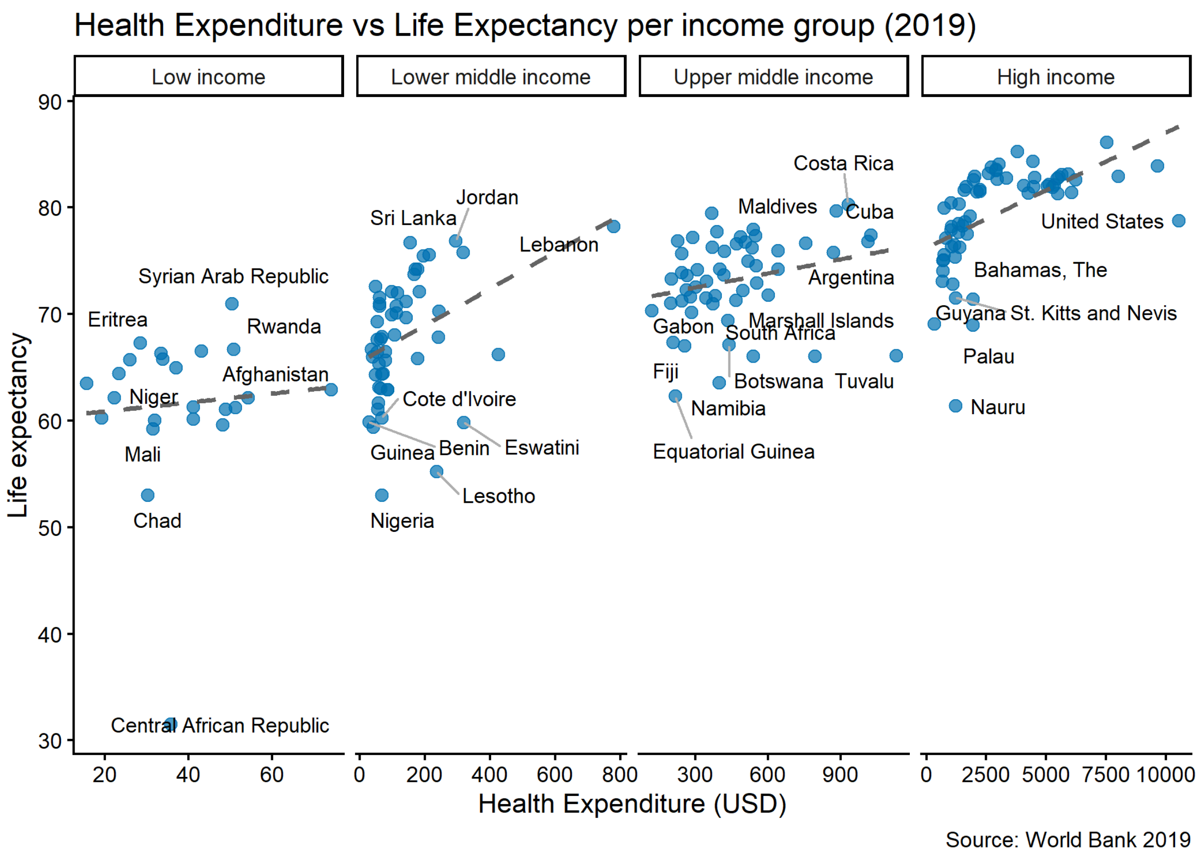
- only oecd so no income groups. Correlation is positive throughout.
- Mostly flat, except for high-income countries. Outliers have a real impact (USA).

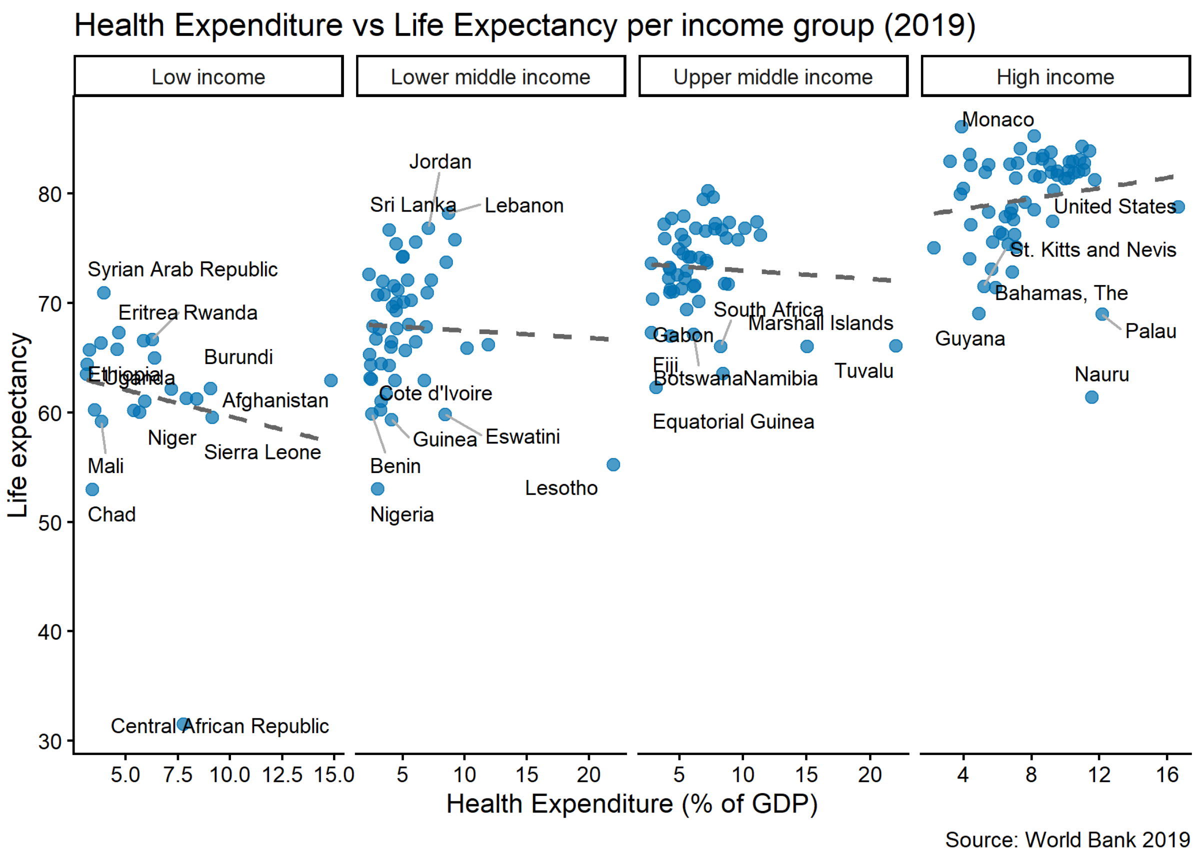
Life expectancy:

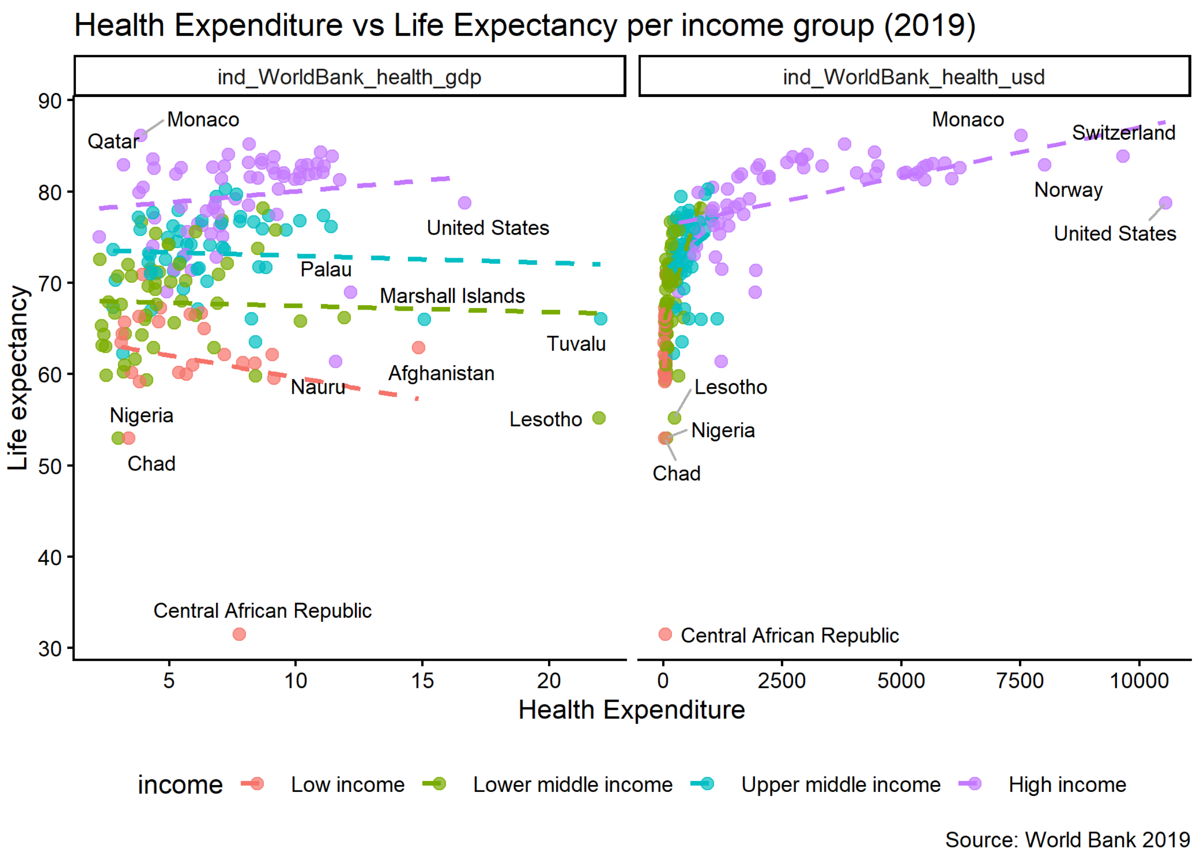
- No, the correlation is not the same across income groups. Our analysis reveals several important patterns:
- High-income countries show stronger correlations between health spending and life expectancy. Strong influence of outliers..
- Lower income countries show negative correlation with share of gdp. In general, lower correlation for low-income countries.
- Clear pattern of diminishing returns in expenses

Limitations:
- The share of GDP measure is relative: low-income countries may have high shares due to small GDPs.
- The per capita USD measure is absolute: that's more strongly correlated with outcomes in cross-country comparisons.




Health Expenditure vs Life Expectancy per income group (2019)
Low income
Lower middle income
Upper middle income
High income
Life expectancy
Health Expenditure (USD)
Source: World Bank 2019
Syrian Arab Republic
Eritrea
Rwanda
Afghanistan
Niger
Mali
Chad
Central African Republic
Jordan
Sri Lanka
Lebanon
Cote d'Ivoire
Guinea
Benin
Eswatini
Lesotho
Nigeria
Costa Rica
Maldives
Cuba
Argentina
Gabon
Marshall Islands
South Africa
Fiji
Botswana
Tuvalu
Namibia
Equatorial Guinea
United States
Bahamas, The
Guyana
St. Kitts and Nevis
Palau
Nauru


Health Expenditure vs Life Expectancy per income group (2019)
Low income
Lower middle income
Upper middle income
High income
Life expectancy
Health Expenditure (% of GDP)
Source: World Bank 2019
Syrian Arab Republic
Eritrea
Rwanda
Burundi
Ethiopia
Uganda
Afghanistan
Niger
Sierra Leone
Mali
Chad
Central African Republic
Jordan
Sri Lanka
Lebanon
Cote d'Ivoire
Guinea
Eswatini
Benin
Lesotho
Nigeria
South Africa
Marshall Islands
Gabon
Fiji
Botswana
Namibia
Tuvalu
Equatorial Guinea
Monaco
United States
St. Kitts and Nevis
Bahamas, The
Guyana
Palau
Nauru




```
# Calculate correlations by income group for 2019
correlation_results_2019 <- df_final |>
  filter(year %in% c(2005, 2019),
         income %in% income_groups) |>
  group_by(income, year) |>
  summarise(
    n_countries = n(),
    cor_usd_life = cor(ind_WorldBank_health_usd, ind_WorldBank_lifeexpectancy,
use = "complete.obs"),
    cor_gdp_life = cor(ind_WorldBank_health_gdp, ind_WorldBank_lifeexpectancy,
use = "complete.obs")
  ) |>
  arrange(income)

print("Correlation between health expenditure and life expectancy by income
group (2019):")
```

```
[1] "Correlation between health expenditure and life expectancy by income
group (2019):"
```

```
print(correlation_results_2019)
```

```
# A tibble: 8 × 5
# Groups:   income [4]
  income               year n_countries cor_usd_life cor_gdp_life
  <fct>               <int>       <int>        <dbl>        <dbl>
1 Low income           2005          23       0.664       0.00853
2 Low income           2019          23       0.0750     -0.177
3 Lower middle income  2005          51       0.270       0.194
4 Lower middle income  2019          51       0.388      -0.0391
5 Upper middle income  2005          53      -0.0126     -0.0256
6 Upper middle income  2019          53       0.250      -0.0598
7 High income          2005          62       0.623       0.176
8 High income          2019          62       0.544       0.139
```

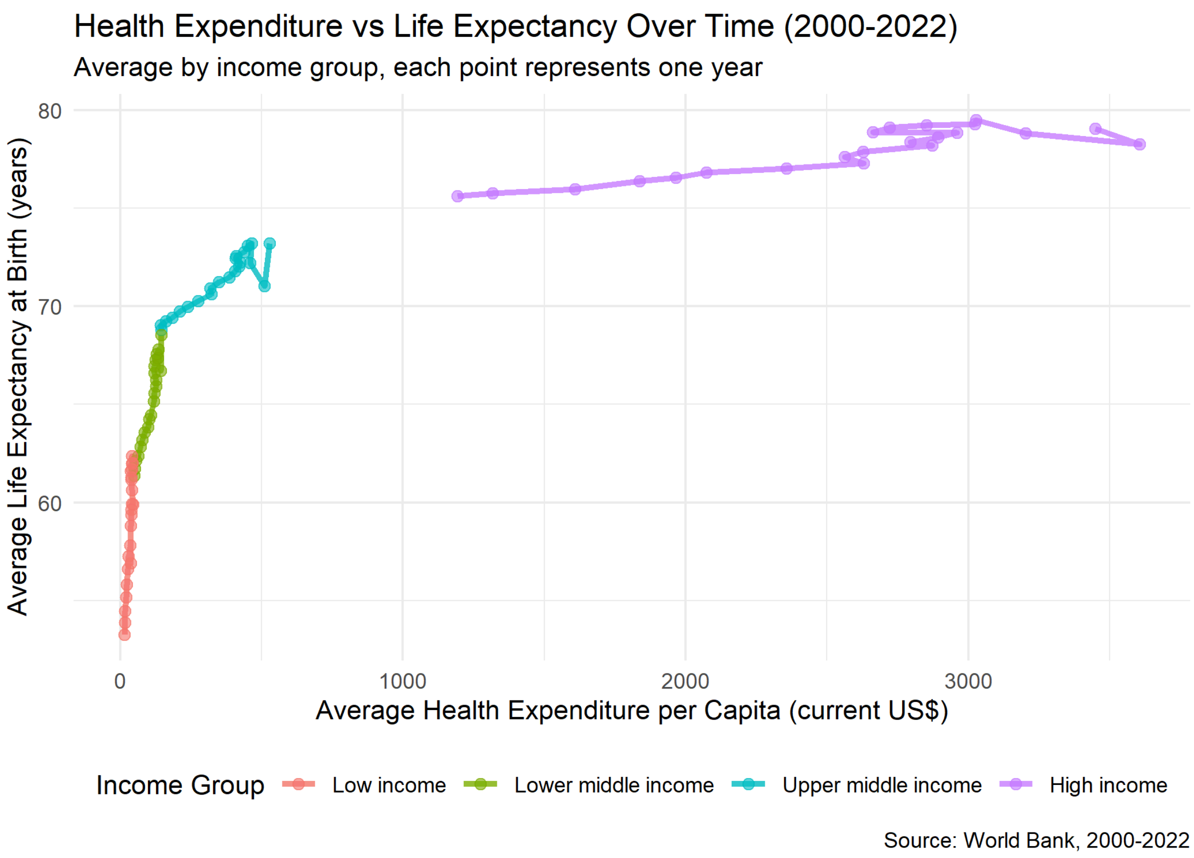
```
# Or per group (to avoid interaction terms)
df_final_income <- split(df_final, df_final$income)

lapply(df_final_income, function(x){
  lm(ind_WorldBank_lifeexpectancy ~ ind_WorldBank_health_usd +
                   factor(year),
                   data = x) |>
    summary()
})


# In gdp
lapply(df_final_income, function(x){
  lm(ind_WorldBank_lifeexpectancy ~ ind_WorldBank_health_gdp +
                   factor(year),
                   data = x) |>
    summary()
})
```

2.6.2.5 Final graph

One nice way to wrap up the whole story of hypothesis 3 in a graph is to show the evolution of life expectancy over time for countries in different income brackets. Plot the countries per group with, on the y axis the life expectancy and on the x axis the expenditures in USD. Aggregate the expenditures per year and group income. Briefly comment. [1pt]

```
# Time series analysis by income group
df_income_time <- df_final |>
  filter(income %in% income_groups, year > 2000) |>
  group_by(year, income) |>
  summarise(
    ind_WorldBank_health_usd = mean(ind_WorldBank_health_usd, na.rm = TRUE),
```

```
    mean_life_expectancy = mean(ind_WorldBank_lifeexpectancy, na.rm = TRUE),
    .groups = "drop"
  )

# Plot the relationship over time
df_income_time |>
  ggplot(aes(x = ind_WorldBank_health_usd, y = mean_life_expectancy, color =
income)) +
  geom_path(alpha = 0.8, size = 1.2) +
  geom_point(alpha = 0.6, size = 2) +
  labs(
    title = "Health Expenditure vs Life Expectancy Over Time (2000-2022)",
    subtitle = "Average by income group, each point represents one year",
    x = "Average Health Expenditure per Capita (current US$)",
    y = "Average Life Expectancy at Birth (years)",
    color = "Income Group",
    caption = "Source: World Bank, 2000-2022"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```



2.7 Wrap up

2.7.1 Your data story [Bonus reflection - perfect for your portfolio!]

If you were to publish this research (like in a blog post, data science portfolio, or LinkedIn article), what would be the compelling story you tell? Summarize your key findings and insights in max. 1000 characters. Feel free to include a visualization that best captures your main message.

Note: This question is not graded but is valuable for developing your communication skills. It works particularly well as a group exercise: have each member brainstorm their own approach, then discuss together to identify the most effective and engaging story.

2.7.2 Further steps [3 pts]

Based on your experience with this analysis, identify three specific ways to enhance this research project. Consider:

- Additional methodological approaches you could explore
- Further data quality checks or robustness tests we did not explore here
- Additional datasets that would strengthen your analysis
- New research questions or mechanisms worth investigating

Note: maximum 50 words in total for this exercise.

- Per gender
- Expenditures in PPP
- Look at variance within group
- …

Bibliography

Anderson, G. F., Reinhardt, U. E., Hussey, P. S., & Petrosyan, V. (2003). It's the prices, stupid: why the United States is so different from other countries. *Health Affairs*, *22*(3), 89–105.